

Insights from aquaporin structures into drug-resistant sleeping sickness

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In this **important** study, the authors set out to determine the molecular interactions between the AQP2 from *Trypanosoma brucei* (TbAQP2) and the trypanocidal drugs pentamidine and melarsoprol to understand how TbAQP2 mutations lead to drug resistance. Using cryo-EM, molecular dynamics simulations, and lysis assays the authors present **convincing** evidence that mutations in TbAQP2 make permeation of trypanocidal drugs energetically less favourable, and that this impacts the ability of drugs to achieve a therapeutic dose. Overall, this data will be of interest for those working on aquaporins, and development of trypanosomiasis drugs as well as drugs targeting aquaporins in general.

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Abstract

Trypanosoma brucei is the causal agent of African trypanosomiasis in humans and animals, the latter resulting in significant negative economic impacts in afflicted areas of the world. Resistance has arisen to the trypanocidal drugs pentamidine and melarsoprol through mutations in the aquaglyceroporin TbAQP2 that prevent their uptake. Here we use cryogenic

electron microscopy to determine the structure of TbAQP2 from *Trypanosoma brucei*, bound to either the substrate glycerol or to the sleeping sickness drugs, pentamidine or melarsoprol. The drugs bind within the AQP2 channel at a site completely overlapping that of glycerol. Mutations leading to a drug-resistant phenotype were found in the channel lining. Molecular dynamics simulations showed the channel can be traversed by pentamidine, with a low energy binding site at the centre of the channel, flanked by regions of high energy association at the extracellular and intracellular ends. Drug-resistant TbAQP2 mutants are still predicted to bind pentamidine, but the much weaker binding in the centre of the channel observed in the MD simulations would be insufficient to compensate for the high energy processes of ingress and egress, hence impairing transport at pharmacologically relevant concentrations. The structures of drug-bound TbAQP2 represent a novel paradigm for drug-transporter interactions that could provide new mechanisms for targeting drugs into pathogens and human cells.

Introduction

Human African trypanosomiasis (HAT) remains a neglected tropical disease¹. It is transmitted by tsetse flies and, once symptoms are apparent, treatment is necessary to avoid fatality. The infection starts with a haemolymphatic stage that progresses to a late stage infection of the central nervous system, at which point severe neurological symptoms occur with rapid progression to coma and death². Early-stage infections are commonly treated with pentamidine and the late stage with melarsoprol, although for most cases this is now being replaced with fexinidazole³. Animal African trypanosomiasis (AAT) causes billions of dollars in damage to agriculture throughout the tropics^{4,5}. For both HAT and AAT, chemotherapy remains the only control option and most of the medication is old, inadequate and threatened by resistance^{3,4,6,7}.

For most trypanocidal drugs, resistance is associated with mutations in the membrane transporters through which the drugs enter the trypanosome⁸. The uptake of pentamidine and melarsoprol was initially attributed to a High Affinity Pentamidine Transporter (HAPT)^{9,10}, which was subsequently identified as the aquaglyceroporin TbAQP2^{11,12}. Deletions and chimeric rearrangements between TbAQP2 and the adjacent gene TbAQP3 were found to be responsible for melarsoprol-pentamidine cross-resistance (MPXR; **Fig. S1**)^{6,12–15}. To understand the differences between TbAQP2 (water/glycerol/drug transporter) and TbAQP3 (classical aquaglyceroporin transporting water/glycerol¹⁶ but not pentamidine or melarsoprol^{11,12,17}), mutants were made exchanging amino acids between the proteins¹⁷. Mutation of TbAQP2 to add any of the TbAQP3 selectivity filter residues Tyr250, Trp102 or Arg256 resulted in inhibition of melarsoprol and pentamidine uptake, whereas TbAQP3 containing all three of their TbAQP2 counterparts (Leu250, Ile102, Leu256) permitted the uptake of pentamidine. However, the molecular details of the interactions between TbAQP2 and drugs as well as the resistance mechanism remained unclear, with alternative models proposing that TbAQP2 acts as a channel^{8,17,18} or receptor¹⁹ for the drugs. To resolve this quandary and to resolve the molecular interactions underlying drug uptake by aquaporins, we determined the structures of TbAQP2 bound to either melarsoprol, pentamidine or the native substrate, glycerol.

Cryo-EM structures of TbAQP2 and drug binding

TbAQP2 was expressed using the baculovirus expression system in insect cells and purified in the presence of either a drug (melarsoprol or pentamidine) or the substrate glycerol (see Methods, **Fig. S2a,b**). Structures of TbAQP2 bound to either glycerol, melarsoprol, or pentamidine were determined by electron cryogenic microscopy (cryo-EM) to overall resolutions of 3.2 Å, 3.2 Å and 3.7 Å, respectively (Figs. S2c, S3 and S4a-c). The resolution was sufficient to see the majority of the side chains and the bound drug or glycerol (Figs S5-S8). The cryo-EM structures of TbAQP2 (excluding the drugs/substrates) were virtually identical (rmsd 0.2 Å) and represented the

aquaporin tetramer (**Fig. 1a,b**) that is typical for this family of channels²⁰. The tetramer contained four channels, each consisting of a single TbAQP2 polypeptide folded into a bundle of eight α -helices (**Fig. 1b**), two of which (helices H3 and H7) penetrated the plasma membrane only partially, as is typical for aquaporins²⁰, the others crossing the membrane fully.

Melarsoprol and pentamidine are bound in a channel formed by the TbAQP2 protomer, completely overlapping the binding site for the substrate glycerol (**Fig. 1c,d**). The densities of the substrates were distinct from one another and are consistent with the structures of the substrates (Figs. S7 and S8). The position of drug binding is consistent with a hotspot analysis performed on the structures (see Methods and **Fig. S2d,e**). This analysis defines likely regions where molecules may bind based on *in silico* docking scores of small molecule fragments docking to a model of TbAQP2²¹ (see Methods). Binding of the two glycerol molecules is mediated by a total of six hydrogen bonds, four to backbone carbonyl groups (residues Gly126, Gly127, His128, Leu129) and two to the side chain of His128 (**Fig. 2a**). These are on a polar strip of residues running down the channel, that also includes Val133, which makes van der Waals contacts with the glycerol molecule closest to the cytoplasm. The remainder of the contacts are van der Waals interactions (Val222, Phe226, Ile241, Val245) on the opposite side of the channel. It is striking that all eleven amino acid residues within 3.9 Å of the two glycerol molecules also make contacts to both pentamidine and melarsoprol (**Fig. 2a**). Whereas the TbAQP2-glycerol contacts are equally polar and non-polar, van der Waals interactions dominate contacts between the drugs and TbAQP2. The exceptions are a hydrogen bond between the backbone carbonyl of Ala259 and both drugs, and a backbone carbonyl hydrogen bond between Gly126 and melarsoprol that is also seen between glycerol and TbAQP2 (**Fig. 2a**). It is also remarkable that an overlay between the three different structures shows a very close alignment between glycerol, pentamidine and melarsoprol (**Fig. 2b**) which is consistent with the channel structure being virtually identical regardless of whether a drug or glycerol is bound. The size of the channel with pentamidine bound is very similar to the glycerol-bound *Plasmodium falciparum* AQP structure (**Fig. S2f**). The close fit of the drugs within the channel is consistent with the extremely limited modification of the drugs that is possible whilst retaining the ability to pass through the pore, including modifications that reduce flexibility¹⁷. Diamidines with fixed-angle structures such as diminazene and furamidine are not substrates for TbAQP2 but are exclusively accumulated through aminopurine transporter TbAT1^{22,23}. A number of amino acid residues interact only with pentamidine and/or melarsoprol and not with glycerol (Leu84, Ile110, Val114, Leu122, Leu218, Ala259, Asn261, Met260, Leu264), which is a result of the drugs being longer and extending further along the channel towards the extracellular surface when compared with glycerol (**Figs. 1d** and **2b**). Towards the intracellular surface, only Leu122 makes contacts to the drugs and not to glycerol. In all, 19 residues in TbAQP2 interact with pentamidine, compared with 18 residues interacting with melarsoprol and 11 to the two molecules of glycerol.

The structure of AQP2 from *Plasmodium falciparum* (PfAQP)²⁴ with glycerol bound is the closest structural homologue to TbAQP2 available (rmsd 0.8 Å, PDB ID 3CO2, 33% amino acid sequence identity). The overall arrangement of α -helices in TbAQP2 is identical to that in PfAQP (**Fig. 3a-d**) and other aquaglyceroporins whose structures have been determined (human AQP7²⁵, human AQP10²⁶ and *Escherichia coli* GlpF²⁷). Despite the low identity in amino acid sequence, there are striking similarities between the structures of TbAQP2 and PfAQP in the conserved NPA/NPA motif and the positions of bound glycerol molecules in the channel. The NPA/NPA motif is found at the ends of the half-helices H3 and H7, with TbAQP2 and PfAQP similarly differing from the canonical sequences (NPS/NSA in TbAQP2 and NPS/NLA in PfAQP; **Fig. 3a,b**). Superposition of TbAQP2 and PfAQP shows the position and rotamer of Asn-Pro-Ser in the NPS motif and both Asn and Ala in the NSA/NLA motifs are identical (**Fig. 3c**). Some of the network of hydrogen bonds that maintain the structure of this region are also conserved, such as the hydrogen bonds in TbAQP2 between Asn130 and the backbone carbonyl of 260 and the backbone amine of Ala132, the hydrogen bond between Ser263 and Gln155 and the hydrogen bond between Asn261 and the backbone amide of Ser263. However, both Asn residues in the NPS/NLA motif of PfAQP form

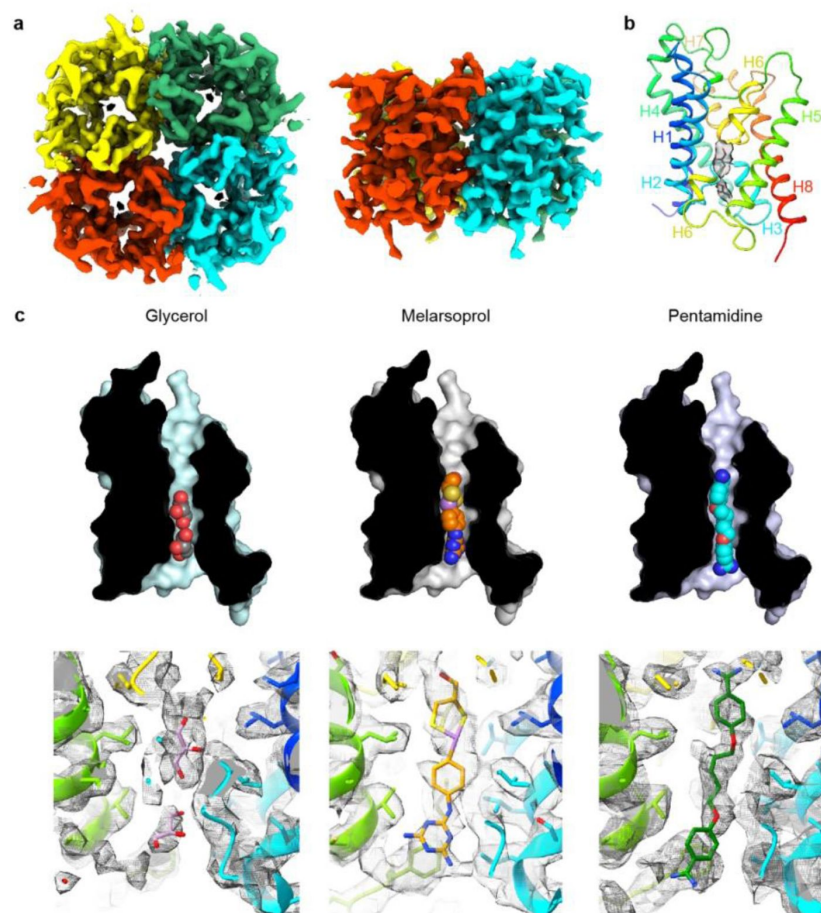


Fig. 1

Cryo-EM structures of TbAQP2 bound to either glycerol, melarsoprol or pentamidine.

a, Overall structure of the TbAQP2 tetramer viewed either from the extracellular surface or within the membrane plane. **b**, Structure of protomer A of TbAQP2 viewed as a cartoon (rainbow colouration) with glycerol (sticks) bound. The cryo-EM density for glycerol is shown as a grey surface. **c**, Cut-away views of the channel in each of the TbAQP2 structures showing bound substrates and drugs (spheres) with atoms coloured according to type: red, oxygen; yellow, sulphur; blue, nitrogen; purple, arsenic; carbon, grey (glycerol), orange (melarsoprol) or cyan (pentamidine). **d**, Cryo-EM densities (grey surface) for glycerol, melarsoprol and pentamidine in their respective structures. See Figures S7 and S8 for different views of the substrates and comparisons between densities.

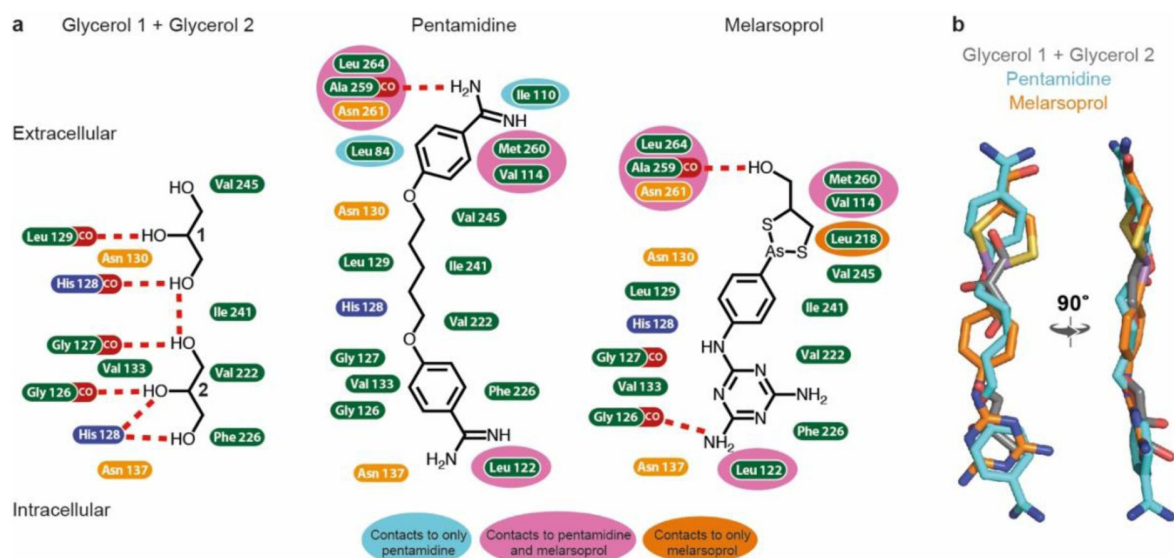


Fig. 2

Interactions between TbAQP2 and bound substrates and drugs.

a, Amino acid residues containing atoms ≤ 3.9 Å from substrate or drug model in the cryo-EM structures are depicted and coloured according to the chemistry of the side chain: hydrophobic, green, positively charged, blue; polar, orange. Interactions with a backbone carbonyl group is shown as CO (red) and potential hydrogen bonds are shown as red dashed lines. Residues highlighted in an additional colour (grey, pale blue, pink or dark orange) make contacts in two or less of the structures (as indicated on the figure), whilst those without additional highlighting make contacts in all three structures. **b**, Protomer A from each structure was aligned and the positions of the two glycerol molecules (grey), melarsoprol (orange) and pentamidine (cyan) are depicted.

hydrogen bonds with one of the glycerol molecules in the channel, whereas the Asn residues in the NPS/NSA motif in TbAQP2 form weak van der Waals interactions with the glycerol molecules, and Asn261 also makes a hydrogen bond to the backbone carbonyl of Leu129. The position of the intracellular loop between H2 and H3 is also conserved between TbAQP2 and PfAQP ([Fig. 3c](#)) with the backbone carbonyls of residues 126, 127 and 128 aligning and all forming hydrogen bonds in both channels to glycerol molecules. The conservation in structure is reflected in the conserved position of the glycerol molecules in the intracellular half of the channel in TbAQP2 and PfAQP, although the orientation of the hydroxyl groups of the glycerol molecules differs. A divergence in the position of the extracellular loop between H6 and H7 in TbAQP2 and PfAQP ([Fig. 3c](#)) leads to a wider channel in TbAQP2 and no ordered glycerol molecules are observed in the channel at this point, unlike in other aquaglyceroporins ([Fig. 3d](#)). Another difference between the transport of glycerol by TbAQP2 and PfAQP is that there is a water molecule coordinated between each glycerol molecule in PfAQP, whereas in TbAQP2 the glycerol molecules interact directly. Any role of water molecules in the transport of glycerol by TbAQP2 remains to be elucidated as the resolution of the current cryo-EM structure is insufficient for resolving them.

Molecular dynamics simulations show impaired pentamidine transport in mutants

To assess the dynamic properties of the structurally-resolved pentamidine binding pose, all-atom molecular dynamics (MD) simulations were performed on TbAQP2 in a tetrameric state (see Methods). In these simulations, both the protein and pentamidine are very stable (protein RMSD = 0.17 ± 0.01 nm and pentamidine RMSD = 0.16 ± 0.03 nm, see [Fig. S9a,e](#)) and pentamidine binding prevents water flux through the channel ([Fig. 4a,b](#)). The stability of the pentamidine is enabled by a number of extremely high occupancy interactions with specific residues lining the TbAQP2 pore ([Fig. S9b](#)). Hydrogen bond analysis reveals around 23 hydrogen bonds may be made between pentamidine and the channel ([Fig. S10a,b](#)). In contrast, water shows a different pattern of high occupancy interactions in the pore, with MD snapshots highlighting several key positions of where waters interact with the channel as they flow through ([Fig. S10c](#)). The presence of pentamidine both severely reduces the number of waters in the channel and slightly impacts their orientation ([Fig. S10d,e](#)). It was not possible to perform MD simulations of TbAQP2 in the presence of melarsoprol, because the presence of an arsenic atom causes force field parameterization issues.

We then explored the translocation of pentamidine through the TbAQP2 channel. The positively charged state of pentamidine suggests that it might be translocated by the $\Delta\Psi$ component of the plasma membrane potential. To test this, we applied a z-dimension electric field to simulations containing monomeric TbAQP2 in a membrane, using either a positive field (the opposite direction to the physiological membrane) or a physiologically-correct negative field polarity. In all simulations, the resulting potential causes the pentamidine to translocate through TbAQP2, and in some simulations (1 out of 3 repeats for each condition) the pentamidine fully leaves the channel ([Fig. 4c](#)). For the physiologically correct negative field, the pentamidine moves downwards ([Fig. 4c](#)), suggesting that the membrane potential is able to import pentamidine into the cell, consistent with experimental evidence that TbAQP2-mediated pentamidine uptake is sensitive to ionophores that depolarise the plasma membrane.^{10,17}

Drug resistant mutations in TbAQP2

T. brucei contains two related aquaglyceroporins, TbAQP2 and TbAQP3, with only TbAQP2 being able to transport pentamidine and melarsoprol into the trypanosome.^{11,12} Drug resistance has arisen through recombination between the two genes and the expression of TbAQP2-AQP3 chimeras.^{13–15} There are 68 amino acid residues that differ between TbAQP2 and TbAQP3 and the drug-resistant chimeras can contain 20–43 differences to TbAQP2 ([Fig. S1](#)). A number of these residues have been mutated individually in attempts to identify key residues involved in

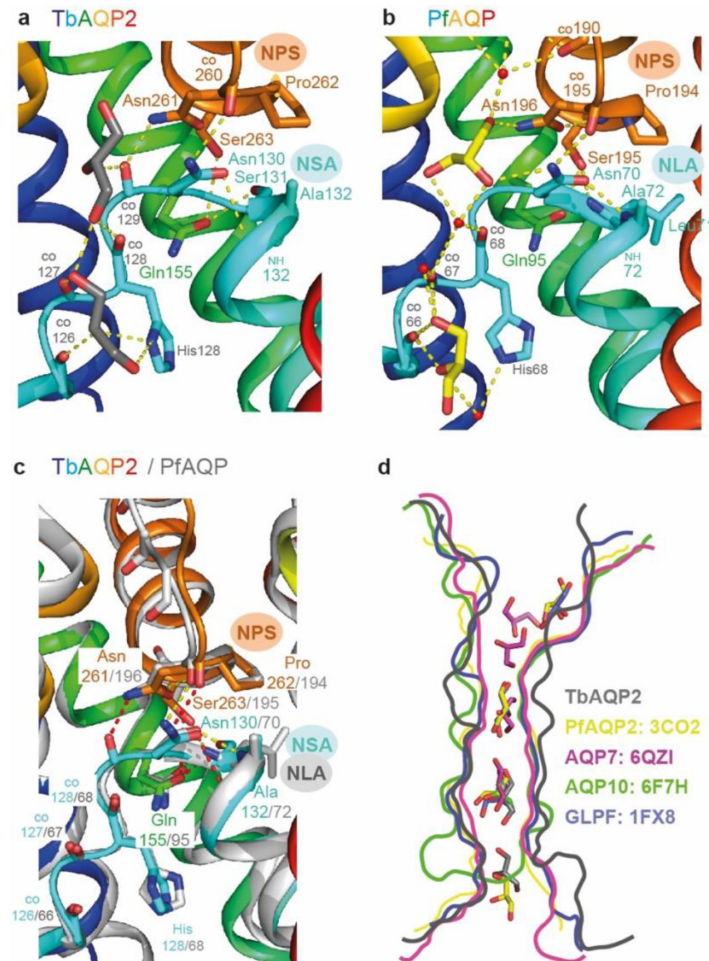


Fig. 3

Comparison between conserved glycerol-binding regions of TbAQP2 and PfAQP.

a, The structure of TbAQP2 is depicted (rainbow colouration) with residues shown that make hydrogen bonds (yellow dashed lines) to either glycerol (grey) or in the NPS/NSA motifs. CO, backbone carbonyl groups involved in hydrogen bond formation. **b**, The structure of PfAQP2 is depicted (rainbow colouration) with residues shown that make hydrogen bonds (yellow dashed lines) to either glycerol (yellow sticks) or water (red spheres), or in the NPS/NLA motifs. CO, backbone carbonyl groups involved in hydrogen bond formation. **c**, Structures of TbAQP2 (rainbow colouration) and PfAQP are superimposed to highlight structural conservation of the NPS and NSA/NLA motifs. **d**, Outline of channel cross-sections from aquaglyceroporins containing bound glycerol molecules (sticks): TbAQP2 (grey); PfAQP (yellow; PDB ID 3CO2); human AQP7 (magenta, PDB ID 6QZI); human AQP10 (green, PDB ID 6F7H); *Escherichia coli* GlpF (purple, PDB ID 1FX8).

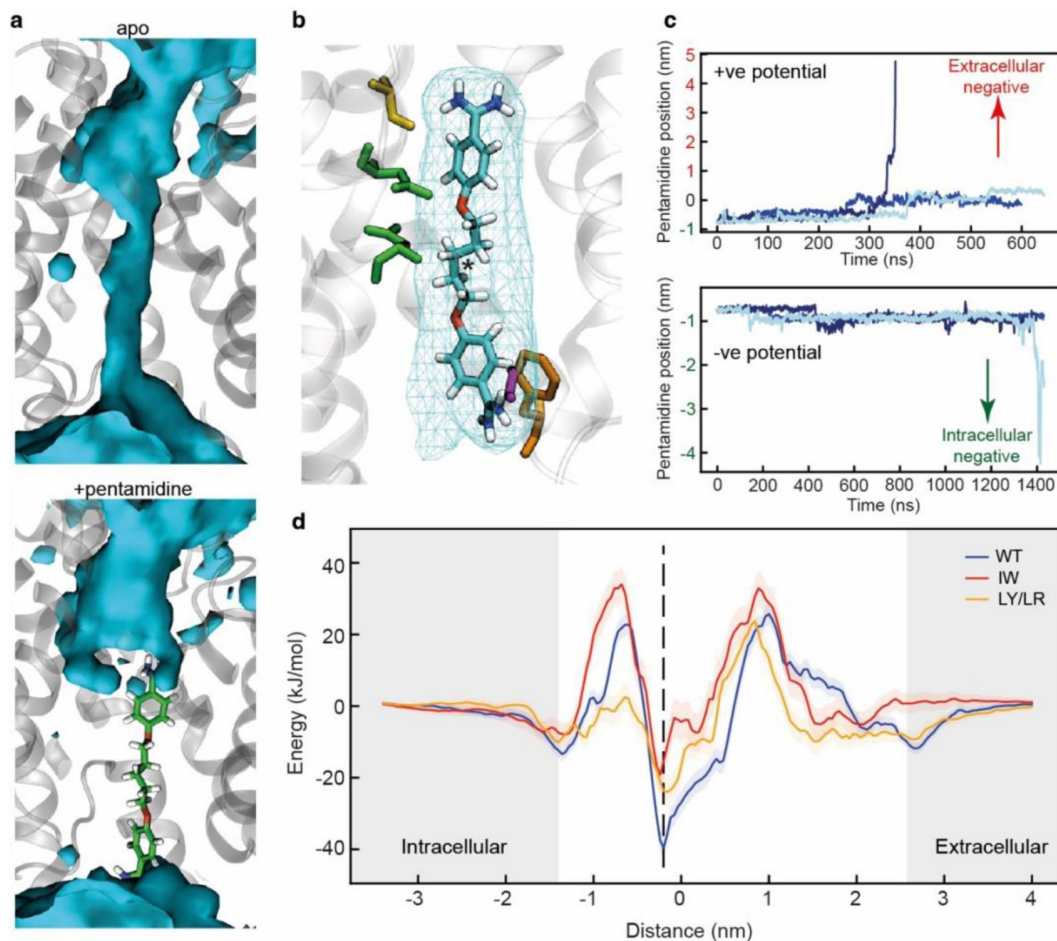


Fig. 4

Molecular dynamics simulations of pentamidine-bound TbAQP2.

a, Density of water molecules (blue) through the TbAQP2 central cavity from molecular dynamics. Bound pentamidine (sticks, green) abolishes the ability of TbAQP2 to transport water molecules. **b**, View of pentamidine (cyan, carbon; red, oxygen; blue, nitrogen; white, hydrogen) bound to TbAQP2. Cyan mesh shows the density of the molecule across the MD simulation, and the asterisk shows the position of the centre of mass (COM). **c**, Upon application of a membrane potential, the pentamidine position as defined by the COM moves along the z-dimension in relation to the COM of the channel, with three independent repeats shown in different shades of blue. The bottom graph is for the potential in the physiological direction (negative intracellular). **d**, Energy landscapes for pentamidine through the TbAQP2 central cavity as calculated using umbrella sampling. Separate calculations were made for monomeric WT TbAQP2 and TbAQP2 with I110W (IW) or L258Y/L264R (LY/LR) mutations. Each trace is built from 167 x 40 ns windows, with the histogram overlap and convergence plotted in **Fig. S12a,b**. The position of the membrane phosphates is shown as grey bars, and the structural binding pose is shown as a dotted line.

drug translocation and thus potentially resistance. At the intramembrane ends of the two transmembrane half-helices H3 and H7 are the aforementioned NSA/NPS motifs, Asn130-Ser131-Ala132 and Asn261-Pro262-Ser263, respectively (**Fig. 3a**). In aquaporins such as TbAQP3 these conserved residues are the canonical NPA/NPA motifs, and mutations were made (S131P+S263A) to convert the NSA/NPS motif in TbAQP2 to NPA/NPA. This resulted in a 95% reduction in pentamidine transport by the mutant, although this was still measurably higher than that observed in a TbAQP2 knockout cell line¹⁷ and a similar reduction in the transport of the melarsoprol analogue cymelarsan (**Fig. S11**). The cryo-EM structures show that both Ser131 and Ser263 are critical in stabilising this region of the transporter through specific hydrogen bonds with Gln155 and the backbone amine group of Ala132 (**Fig. 3a**) and do not make any direct interactions either to the drugs or the glycerol molecules. Thus, the decrease in pentamidine uptake observed in the S131P+S263A TbAQP2 mutant is probably more complex than previously thought, as it seems unlikely to have arisen through simple changes in transporter-drug interactions or a narrowing of the channel.

Based on a previous modelling study, several other mutations were predicted to potentially affect drug uptake in the TbAQP2-TbAQP3 chimeras¹⁷. Based on the cryo-EM structures, the mutated residues fall into three distinct groups: those not in the channel (I190T, W192G), those lining the channel and within van der Waals distance of pentamidine (I110W, L258Y, L264R) and those lining the channel, >3.9 Å from pentamidine and observed to make contacts to pentamidine during MD simulations (L84W/M, L118W/M, L218W/M). In all instances, there was a >90% decrease in pentamidine uptake) and in many cases no residual pentamidine transport was detected (e.g. I110W, L264R, L84W)¹⁷. To understand the molecular basis for the dramatic decreases observed in pentamidine transport, we performed molecular dynamics simulations on a subset of the mutants.

Three mutations were chosen for molecular dynamics simulations (I110W, L258Y, L264R), all of which were defective in uptake of cymelarsan (**Fig. S11**) as well as pentamidine¹⁷. These three residues all make high occupancy contacts to pentamidine in the MD simulations (**Fig. S9b**) and residues Ile110 and Leu264 make van der Waals contacts with pentamidine in the cryo-EM structure (Leu258 is 5.5 Å away from pentamidine). The double mutant L258Y+L264R was used in the MD simulations as this is found in a number of drug-resistant *T. brucei* strains such as P1000 and Lister 427MR¹². The mutant I110W is found in the *T. brucei* drug resistant strain R25²⁸, but in the absence of the L258Y+L264R double mutation (**Fig S1**). Therefore, we performed atomistic MD simulations of monomeric TbAQP2 with either the I110W (hereafter IW) or L258Y+L264R (hereafter LY/LR) mutations present, as well as with wild-type monomeric TbAQP2. Each protein was simulated both with pentamidine bound and removed (apo). The mutations do not affect TbAQP2 stability (**Fig. S7e**), but do destabilise pentamidine binding, causing it to shift away from the initial binding pose along the z-axis (**Fig. S9c**), suggesting a lowered affinity for the structural pose. The pore radii are unchanged by the mutations (**Fig. S9d**), meaning that this effect is likely to be a loss of specific interactions rather than a more general effect on protein structure.

We next computed energy landscapes for pentamidine moving along the z-axis through the channel using potential of mean force (PMF) calculations with umbrella sampling. The data for WT TbAQP2 reveals a clear energy well for the structurally resolved pentamidine at -0.2 nm (**Fig. 4d** and **Fig. S11c**). The depth of this well (-40 kJ/mol) suggests a very high affinity binding site, consistent with previous kinetic analyses^{9,10,29}, and is much higher than previously investigated docked poses¹⁷ of pentamidine in models of TbAQP2. Notably, this energy well is much shallower for both the IW (-20 kJ/mol) and LY/LR mutants (-25 kJ/mol), representing a relative reduction in binding likelihood of about 2,800-fold and 400-fold, respectively. In biochemical terms, these changes would result in a K_m shift from ~50 nM to 50 or 300 μM, respectively, all but abolishing uptake at pharmacologically relevant concentrations; the experimental K_m for pentamidine on wild type TbAQP2 is 36 nM¹⁰. The reduction in pentamidine

binding likelihood is consistent with the increased z-axis dynamics seen in **Fig. S9c**. Flanking the central energy well are large energy barriers at about -0.7 nm and 1 nm (**Fig. 4d**), similar to those seen for urea translocation in HsAQP1³⁰. These would presumably slow pentamidine entry into and through the channel, and previous studies did reveal a slow transport rate for pentamidine^{10,29}. We note that previous studies using these approaches saw energy barriers of a similar size, and that these are reduced in the presence of a membrane voltage^{17,31}. The energy barriers are highest in the IW mutant, and interestingly the inner (-0.7 nm) barrier is lower in the LY/LR mutant (**Fig. S12e**). In all cases, especially WT, small energy minima are observed at about -1.35 nm and +2.7 nm; these represent the drug binding at the entrance and exit of the channel (**Fig. S12d**). The differences in landscape between the WT and mutants helps explain how these mutations confer a resistance to pentamidine, mostly through reduced binding affinity.

Discussion

It has only recently become accepted that the aquaglyceroporin AQP2 in *T. brucei* is acting as a *bona fide* transporter of pentamidine and melarsoprol that is essential for translocation of the drugs into the cell¹⁷. The cryo-EM structures of TbAQP2 identify the binding site for melarsoprol and pentamidine in a wide channel that has, at least for part of the trajectory, similar dimensions to channels in other aquaglyceroporins (**Fig. 3d**). The overall similarity between TbAQP2 and the structures of other aquaglyceroporins suggests a similarity in the mechanism of glycerol transport, despite differences in the amino acid residues lining the channel. It also suggests that there is scope for pharmacological targeting of other aquaglyceroporins, either for uptake of cytotoxic drugs or inhibition of glycerol uptake. Leishmaniasis, for instance, is another neglected tropical disease, caused by parasitic protozoa of the genus *Leishmania*; it is commonly treated by pentavalent antimonials meglumine antimoniate and sodium stibogluconate. The antimony that is released in the acidic parasitophorous vacuole enters the parasite through its AQP1 aquaporin³². Widespread antimonial drug resistance in *Leishmania donovani* in India has been linked to a frameshift mutation in LdAQP1³³. In another example, *Plasmodium spp.* contain a single aquaglyceroporin and studies in mouse models have shown that a knockout of the aquaglyceroporin leads to a significant loss of virulence and reduction in erythrocyte infectivity by merozoites³⁴, suggesting that a specific AQP inhibitor could be part of antimalarial treatments. Interestingly, pentamidine is a known antimalarial³⁵ and although a substrate of the *P. falciparum* choline transporter, does not act through inhibition of that carrier³⁶; inhibition/permeation of PfAQP by pentamidine has not been investigated to date. The structure determination of TbAQP2 highlights new opportunities for drug development targeting the aquaporin family to treat a number of parasitic diseases whose current treatments often suffer from severe side effects and growing problems in drug resistance. A thorough understanding of the interaction of extended, flexible drugs like pentamidine and melarsoprol with AQP channel linings could contribute to the design of new substrates able to penetrate not just protozoan but also human AQPs.

After this work was completed, a paper was published describing the structures of TbAQP2 bound to pentamidine and melarsoprol³¹. The structures of TbAQP2 are very similar to the ones described here (RMSD 0.6 Å), despite the environments of the purified channels being significantly different. In the work described here, TbAQP2 was purified in detergent and then incorporated into saposin A (SapA)³⁷ lipid nanoparticles (see Methods), whereas Chen *et al.*³¹ incorporated TbAQP2 into lipid nanodiscs formed from a helical membrane scaffold protein (MSP)^{38,39}. Lipid nanodiscs and nanoparticles are considered better mimetics of a lipid bilayer than detergent micelles, although biophysical studies indicate that there are significant differences between these lipidic structures and a biological membrane^{40,41}. Comparison of the pentamidine in the different structures indicates good agreement in its position within the channel, although the pose of the drug does differ slightly (**Fig. 5a**). In contrast, there is a significant difference in the position and pose of melarsoprol between the different structures (**Fig. 5b**). Melarsoprol in the

TbAQP2 structure from Chen *et al.*³¹ is shifted further towards the extracellular surface of the channel and is rotated approximately 180° about an axis in the membrane plane and centred on the arsenic atom (**Fig 5b**). This unexpected difference in orientation of melarsoprol in the two structures may perhaps not be so surprising as could be initially thought. The two structures represent the thermodynamically most stable states under the buffer conditions and detergent/lipid nanoparticles used for protein purification, which were different. In addition, the site of localisation of the drugs in the TbAQP2 channel is not a physiologically relevant ‘binding site’ as used to describe, for example, the orthosteric binding site in G protein coupled receptors where a hormone or neurotransmitter binds. Thus, evolution has not evolved a high-affinity binding site in the TbAQP2 channel and neither were the drugs designed to bind to TbAQP2. The channel merely facilitates the entry of the drugs into the cell through passive diffusion and it is not necessary for its mode of action to bind to TbAQP2 in a specific pose. Indeed, *a priori*, there is no reason to think that melarsoprol has to traverse the channel in any particular orientation, although under cellular conditions where a membrane potential is present it might be anticipated that the melamine moiety may enter the channel first given its net positive charge. Regrettably, in the absence of molecular dynamics simulations (due to the issues with parameterization of the arsenic atom) these ideas have to remain untested.

Methods

Data reporting

Sample size was not predetermined by statistical methods, no randomization was applied to experiments and investigators were not blinded to experimental allocation or outcome assessment.

Cloning and expression of wild-type AQP2

A construct encoding wild-type *T. brucei* AQP2 (residues 1–312) that included a tobacco etch virus (TEV) cleavage site, eGFP, and a decahistidine tag (TbAQP2-TEV-eGFP-His10) was generated and cloned into a pFastBac vector (Thermo Fisher) using standard ligation procedures. Recombinant pFastBac-TbAQP2-TEV-eGFP-His10 vector was transformed into Stellar Competent Cells (Takara) following the Bac-to-Bac Baculovirus Expression System manual (Thermo Fisher). Expression of AQP2 in *Spodoptera frugiperda* 9 (Sf9) cells (Thermo Fisher Scientific) was performed following the Bac-to-Bac Baculovirus Expression System manual (Thermo Fisher). The cells were then collected by centrifugation (10 min, 4°C, 3000 ×g), flash-frozen in liquid nitrogen, and stored at -80 °C until further use.

Purification of AQP2 bound to either glycerol, pentamidine or melarsoprol and reconstitution into saposin-A nanoparticles (SapA-NP)

All steps described below were carried out at 4°C. Sf9 cells were defrosted on ice, resuspended in cold homogenization buffer (100 mM NaCl, 20 mM HEPES pH 7.5, 5% (v/v) glycerol, 2 mM PMSF, 1× Complete Protease Inhibitor Cocktail (Sigma)) and homogenised by applying 20 strokes in a Potter-Elvehjem glass homogenizer on ice. Membranes were solubilised using 1% (w/v) final concentration n-dodecyl-β-D-maltopyranoside (DDM, Anatrace) and a final concentration of 0.03% (w/v) cholesteryl hemisuccinate (CHS, Anatrace) for 4 hours at 4°C on a roller platform. Insoluble material was removed by ultracentrifugation (100,000 ×g, 60 min, 4 °C). The solubilisate was supplemented with 20 mM imidazole (final concentration) and incubated (16 h, 4°C) with 2 ml of Ni²⁺-NTA agarose resin (Qiagen) per litre of Sf9 cell culture. The resin was then applied to a gravity-flow column and washed three times with 20 column volumes washing buffer (100 mM NaCl, 20 mM HEPES pH 7.5, 2% (v/v) glycerol, 2 mM PMSF, 1× Complete Protease Inhibitor Cocktail

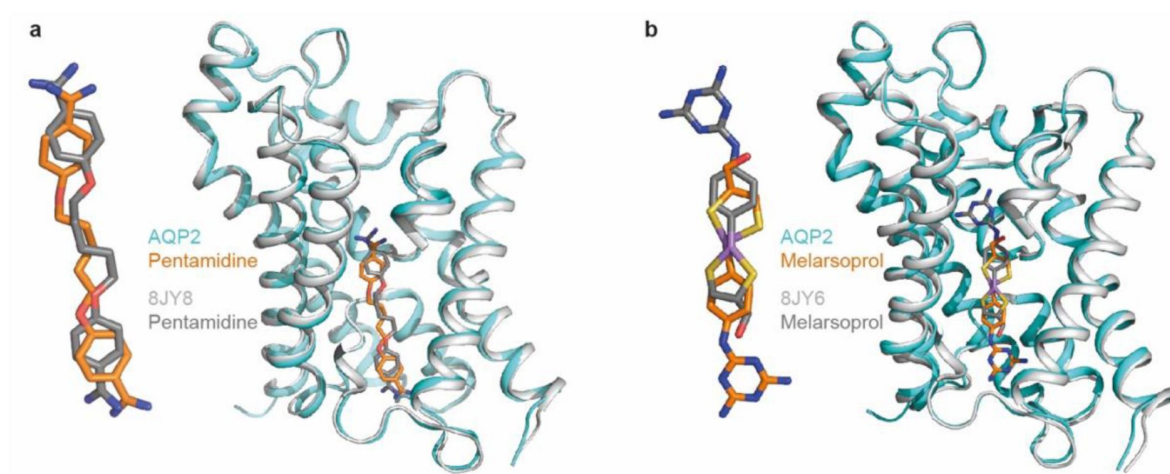


Fig. 5

Comparison between drug-bound AQP2 structures.

Superposition of an AQP2 protomer from this work (cyan) with a protomer determined by Chen *et al.*³¹ (grey), with the positions of drugs shown in stick representation: **a**, pentamidine; **b**, melarsoprol.

(Sigma), 0.05% (w/v) DDM, 0.03% (w/v) CHS, 100 mM imidazole) and was eluted twice with one column volume elution buffer (100 mM NaCl, 20 mM HEPES pH 7.5, 2% (v/v) glycerol, 1× Complete Protease Inhibitor Cocktail (Sigma), 2 mM PMSF, 500 mM imidazole) each. Both eluates were combined and concentrated in an Amicon Ultra 4 mL centrifugal filter (100,000 MWCO, Merck) and aggregates were removed using a syringe-driven filter units (Millex-GV, 0.22 µm, PVDF, 13 mm, Merck). The filtrate was loaded at 0.4 ml/min onto a Superdex Increase S200 10/300 (GE Healthcare) size exclusion chromatography (SEC) column, pre-equilibrated with 0.05% (w/v) DDM, 100 mM NaCl, 20 mM HEPES pH 7.5. The peak fractions containing AQP2 were pooled and concentrated as above. Protein concentration was estimated by the Pierce BCS assay according to the manufacturer's instructions.

SapA-NPs were prepared and stored in SapA-NP buffer consisting of 150 mM NaCl, 50 mM sodium acetate pH 4.8)⁴². Purified TbAQP2 tetramer was combined at a TbAQP2:SapA:lipid molar ratio of 0.1:1:10 and dialysed (20 h, 4°C) against SapA-reconstitution buffer (100 mM NaCl, 20 mM HEPES pH 7.5). SapA-NPs, unincorporated TbAQP2 and SapA were removed by SEC and SapA-TbAQP2-NP tetramer complex was concentrated using Amicon Ultra centrifugal filters (100,000 MWCO; Merck). Melarsoprol or pentamidine was added to the purified SapA-TbAQP2-NP tetramer complex (final concentration 1 mM, 2 h, 4°C) applied to a Superdex Increase S200 10/300 SEC column (GE Healthcare) and fractions containing the complex were concentrated using Amicon Ultra centrifugal filters (100,000 MWCO; Merck) to final concentrations between 0.8 mg/ml and 1.6 mg/ml. Final samples were centrifuged (1 h, 18,000 ×g, 4°C) immediately prior to cryo-EM grid preparation.

Vitrified sample preparation and data collection

Cryo-EM grids were prepared by applying 2 µl of purified SapA-TbAQP2-NP with either glycerol, pentamidine or melarsoprol bound (0.8-1.6 mg/ml) onto either glow-discharged PEGylated UltrAuFoil R1.2/1.3 300 Mesh Gold grids (Agar scientific, No. AGS187-3) or glow-discharged Quantifoil R1.2/1.3 300 Mesh Gold grids (Agar scientific, No. AGS143-8). Grids were blotted with filter paper for 4 s before plunge-freezing in liquid ethane (at -181 °C) using an Thermo Fisher Scientific Vitrobot Mark IV (95% relative humidity, 4°C). All cryo-EM datasets were collected on Titan Krios microscopes (Thermo Fisher Scientific) operating at 300 kV. SapA-TbAQP2-NP with glycerol was collected at the Nanoscience Centre (University of Cambridge) with a Falcon 3 detector to obtain dose-fractioned movies. 2881 movies were recorded with 75,000× magnification at 1.07 Å per pixel using an exposure time of 60 s resulting in a total exposure of (30 e/pixel) and a target defocus range of -1.4 and -3.2 µm. Datasets of SapA-TbAQP2-NP with bound pentamidine or melarsoprol were collected at the cryo-EM facility at the Department of Biochemistry, University of Cambridge; images were recorded using a K2 detector in counting mode, operating with GIF Quantum LS imaging filter (Gatan) with a slit width of 20 eV to obtain dose-fractioned movies with a 100 µm objective aperture. 2400 movies were recorded for SapA-TbAQP2-NP with pentamidine bound and 3000 movies for melarsoprol bound complex. Micrographs for both datasets were recorded at a magnification of 130,000× (1.07 Å per pixel) as dose-fractioned movie frames with an exposure time of 15 s and 12 s, respectively, resulting in a total exposure of 54 e/Å² and 57.5 e/Å² with target defocus ranges of -1.5 and -3.3 µm, and -1.2 and -3 µm, respectively.

Cryo-EM data processing and 3D reconstruction

Cryo-EM data processing was conducted inside the Scipion platform⁴³. Image stacks (2,881 glycerol-bound, 4,086 melarsoprol-bound and 3,262 pentamidine-bound) were subjected to beam-induced motion correction using Relion's implementation of MotionCor2⁴⁴ by dividing each frame into 5 × 5 patches. CTF parameters were estimated from dose-weighted micrographs as follows: estimations made with GCTF⁴⁵ with equiphase averaging and with CTFFIND-4.1^{46,47} were compared using «xmipp-ctf consensus». Outputs from these consensus (micrographs for which the two algorithms agreed in their estimations) undergone an additional round of ctf estimation with «xmipp-ctf estimation»⁴⁸, followed by a second round of xmipp-ctf

consensus for micrograph curation based on defocus, astigmatism, resolution and ice thickness. As a result, the number of micrographs for downstream analysis was 2102, 3586 and 2983 for SapA-TbAQP2-NP bound to either glycerol, melarsoprol or pentamidine, respectively.

Autopicking was performed using crYOLO⁴⁹, that yielded 849,607 particles, 1,927,102 particles and 1,263,534 particles for SapA-TbAQP2-NP bound to either glycerol, melarsoprol or pentamidine, respectively. Particles were extracted in a box-size of 250 px at the original sampling rate of the micrographs (1.07 Å/px) and subjected to several rounds of 2D classifications with cryoSPARC⁵⁰. To further curate the particle set (ligand-free: 100,075; melarsoprol: 202,752; pentamidine: 114,206). Particles were re-extracted with a box of 320 px before 3D classifications.

For the glycerol-bound dataset, an *ab initio* 3D model was generated using stochastic gradient descent algorithm implemented in cryoSPARC, followed by non-uniform refinement⁵¹ with C4 symmetry and a static mask encompassing the transmembrane domains and nanodiscs belt, while excluding the mobile GFP tags. With these settings, a 3.2 Å-resolution map was obtained. For the melarsoprol and pentamidine datasets, six *ab initio* models were generated. Particles from the best classes (melarsoprol: 126,551; pentamidine: 83845) were used for non-uniform-refinement with the same strategy as for the glycerol-bound particles, yielding a 3.2 and 3.7 Å-resolution map, respectively.

The global resolution was calculated following the FSC gold standard threshold of 0.143. The refined maps were sharpened using deepEMhancer⁵² to facilitate model building. Finally, voxel size for melarsoprol and pentamidine was re-adjusted to 1.042 Å/px using Relion post-process⁵³. Local resolution was estimated with MonoRes⁵⁴.

Structure determination and model refinement

The monomer structure of TbAQP2 was predicted using the AlphaFold2 program as a starting point for further refinement⁵⁵. The final cryo-EM map was converted to an mtz file with a range of sharpening and blurring options using CCPEM^{56,57}. EMDA was used for fitting maps⁵⁸. Real-space refinement of the atomic coordinates into the cryo-EM map was performed using Phenix⁵⁹ and Coot software⁶⁰. Finally, models in the asymmetric unit were refined against unsharpened and unweighted half maps using Servalcat Refmac5 pipeline⁶¹ with C4 symmetry constraints. The reference structure restraints were prepared with ProSmart⁶² using AlphaFold2 predicted models from the AlphaFold DB⁶³. The final models were evaluated for geometry, close contacts, and bond parameters using MolProbity⁶⁴. Graphic representations of the fitted coordinates into electron density and the final cryo-EM map were generated using PyMOL⁶⁵ and UCSF ChimeraX^{66,67} packages.

Molecular dynamics simulations

The model for MD simulations was based on the structure of TbAQP2 with pentamidine bound and was built into simulation boxes using CHARMM-GUI^{68,69} (see [Table S2](#) for a list of systems built). Protein atoms were described with the CHARMM36m force field^{70,71}. TbAQP2 was built either as a tetramer or a monomer. Side chain pKas were assessed using propKa3.1 and side chain side charge states were set to their default⁷². The pentamidine molecule used existing parameters available in the CHARMM36 database under the name PNTM with a charge state of +2 to reflect the predicted pKas of >10 for these groups⁷³ and in line with previous MD studies¹⁷. For the ligand-free simulations, the pentamidine molecule was removed prior to simulation.

The proteins were built into membranes comprising 20% cholesterol and 80% POPC, solvated with explicit water molecules using the TIP3P model, and neutralised with K⁺ and Cl⁻ to 150 mM. Box sizes were initially set to 10 × 10 × 10 nm for the tetrameric systems and 6.5 × 6.5 × 10 nm for the monomeric systems. The use of an all-atom fixed charge force field with explicit solvent and membrane is optimal to understanding the protein-drug interactions central to this study. Each

system was minimized and equilibrated according to the standard CHARMM-GUI protocol^{68,69}. Production simulations using random seeds were run in the NPT ensemble with 2 fs time steps, temperatures held at 303.5 K using a velocity-rescale thermostat and a coupling constant of 1 ps, and pressure maintained at 1 bar using a semi-isotropic Parrinello-Rahman pressure coupling with a coupling constant of 5 ps^{74,75}. Short range van der Waals and electrostatics were cut off at 1.2 nm, and the particle mesh Ewald (PME) method was used for long-range Lennard-Jones interactions⁷⁶. Hydrogen bonds were constrained using the LINCS algorithm.

Membrane potential simulations were run using the computational electrophysiology protocol. An electric field of 0.1 V/nm was applied in the z-axis dimension only, to create a membrane potential of about 1 V (see **Fig. S12a**). Note that this is higher than the physiological value of 87.1 ± 2.1 mV at pH 7.3 in bloodstream *T. brucei*⁷⁷ and was chosen to improve the sampling efficiency of the simulations. The protein and lipid molecules were visually confirmed to be unaffected by this voltage, which we quantify using RMSF analysis on pentamidine-contacting residues (**Fig. S12b**).

Potential of mean force (PMF) calculations were built from initial steered MD pulls using an umbrella potential along a coordinate of the COM of the pentamidine molecule and the COM of select residues in TbAQP2 which form the pentamidine binding site (Leu84, Ile110, Val114, Leu122, Gly127, Leu129, Leu218, Val222, Phe226, Leu244, Val245, Ala259, Met260, and Leu264). Pulls were run in the z-axis in either a positive or negative direction, at a rate of 1 nm per ns and with a force constant of 1000 kJ/mol/nm². Snapshots were taken along the coordinates every 0.05 nm, using a script from Dr. Owen Vickery (<https://doi.org/10.5281/zenodo.3592318>), with a total of 167 snapshots to construct the reaction coordinate. End points for the coordinate were determined based on the pentamidine no longer making any contacts (<0.4 nm) with any atoms from the protein or membrane. Simulations were run for 5 ns per window with a 5,000 kJ/mol/nm² umbrella restraint imposed to keep the system in place with respect to the reaction coordinate. An atom in Leu218 was chosen as a reference for the treatment of periodic boundary conditions. Forces were written every 1 ps. PMF profiles were constructed using the weighted histogram analysis method (WHAM)⁷⁸ implemented in *gmx wham*⁷⁹ between -3.5 nm to 4.5 nm of the coordinate and with cyclisation applied. The first 50 ns of each window was discarded as equilibration time and 200 Bayesian bootstraps were used to estimate the PMF error. Adequate coverage of the landscape was assessed based on histogram overlap (**Fig. S10c**), and convergence was determined based on re-running the WHAM calculations on longer windows until the landscapes converged (**Fig. S10d**).

All simulations were run in Gromacs 2020.3 (<https://doi.org/10.5281/zenodo.7323409>)⁸⁰. Data were analysed using Gromacs tools and VMD⁸¹. Pentamidine interactions with TbAQP2 were assessed using PyLipID⁸². Pore radii were assessed using the CHAP package⁸³. Plots were made using Matplotlib⁸⁴ or Prism 9 (GraphPad). MD input and output coordinate files are available at <https://osf.io/zc235/>.

Docking and hotspots experiments

Fragment Hotspot Maps (FHMs) is a powerful tool for predicting the binding hotspots of small molecules on protein surfaces²¹. FHMs are based on the concept that certain chemical fragments are more likely to interact with specific regions of a protein's surface, or 'hotspots', than others. By systematically mapping the interactions between a library of fragment molecules and a target protein, FHMs can identify key hotspots and help guide the design of new small molecules with improved binding affinity and specificity. The development of FHMs has been driven by advances in computational methods for predicting protein-ligand interactions, as well as by improvements in experimental techniques for mapping the 3D structure of protein-ligand complexes. FHMs have been used to guide drug discovery efforts in a variety of therapeutic areas, including oncology, infectious diseases, and neurology.

One key advantage of FHMs is that they can help to identify "druggable" regions of a protein's surface that may be otherwise difficult to target with small molecules. For example, many protein-protein interaction interfaces are large and flat, making it challenging to design small molecules that can disrupt these interactions. FHMs can help to identify the specific regions of the interface that are most important for binding and guide the design of molecules that can effectively target these regions. Another advantage of FHMs is that they can be used to predict the binding modes of small molecules that have not yet been synthesized or tested experimentally. This can be particularly useful in early-stage drug discovery, where computational methods are often used to screen large databases of compounds for potential hits. Overall, FHMs represent a powerful approach for predicting and optimizing small molecule binding to protein surfaces.

We used FHMs program to generate hotspots for AQP2 monomer and probe protein's potential ligand binding sites. The superposition of the final structures and the predicted hotspot maps show a clear overlap between the hotspots and the two drugs (**Fig. S2** [↗](#)).

Lysis assay with arsenical drugs

The effects of cymelarsan and phenylarsine oxide (PAO) on *T. brucei* Lister 427 bloodstream forms was measured as the decrease in cell absorbance at λ 750 nm over time, based on the reduction of cell motility and increased cell lysis leading to reduced light scatter and absorbance in the cuvette⁸⁵ [↗](#), exactly as described⁸⁶ [↗](#). Briefly, 2×10^6 cells in 200 μ L buffer was placed in the wells of a 96-well plate and absorbance recorded in a PHERAstar plate reader (BMG Labtech, Durham, NC, USA) for 10 min before the addition of 20 μ L of 100 μ M of either cymelarsan or PAO in aqueous buffer, or buffer alone.

Supplementary Tables and Figures

TbbAQP2 (+glycerol) MQSQPDNVAYPMELQAVNKDGTVEVRVQGNVDNSSNERWDADVQKHEVAEAQEKPVGGINFWAPR 65

TbbAQP2 (+pentamidine) MQSQPDNVAYPMELQAVNKDGTVEVRVQGNVDNSSNERWDADVQKHEVAEAQEKPVGGINFWAPR 65

TbbAQP2 (+melarsoprol) MQSQPDNVAYPMELQAVNKDGTVEVRVQGNVDNSSNERWDADVQKHEVAEAQEKPVGGINFWAPR 65

TbgAQP2/3 (678-880) MQSQPDNVAYPMELQAVNKDGTVEVRVQGNVDNSSNERWDADVQKHEVAEAQEKPVGGINFWAPR 65

TbbP1000 MQSQPDNVAYPMELQAVNKDGTVEVRVQGNVDNSSNERWDADVQKHEVAEAQEKPVGGINFWAPR 65

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TbbR25 MQSQPDNVAYPMELQAVNKDGTVEVRVQGNDD-SSN-R-----KHEVAEAQEEVPGGINFWAPR 57

TbbAQP3 MQSQPDNVAYPMELQAVNKDGTVEVRVQGNDD-SSN-R-----KHEVAEAQEEVPGGINFWAPR 57

*****:*****

TbbAQP2 (+glycerol) ELRLNYRDYVAEFLGNFVLIYIAKGAVITSLLPDFGLGLTIGIGVAVTMALYVSLGISGGHLN 130

TbbAQP2 (+pentamidine) ELRLNYRDYVAEFLGNFVLIYIAKGAVITSLLPDFGLGLTIGIGVAVTMALYVSLGISGGHLN 130

TbbAQP2 (+melarsoprol) ELRLNYRDYVAEFLGNFVLIYIAKGAVITSLLPDFGLGLTIGIGVAVTMALYVSLGISGGHLN 130

TbgAQP2/3 (678-880) ELRLNYRDYVAEFLGNFVLIYIAKGAVITSLLPDFGLGLTIGIGVAVTMALYVSLGISGGHLN 130

TbbP1000 ELRLNYRDYVAEFLGNFVLIYIAKGAVITSLLPDFGLGLTIGIGVAVTMALYVSLGISGGHLN 130

Tbb427MR ELRLNYRDYVAEFLGNFVLIYIAKGAVITSLLPDFGLGLTIGIGVAVTMALYVSLGISGGHLN 130

TbbR25 ELRLNYRDYMGELLGT FVLLFMGNGVATVIIDGKLGFLSITLWGI AVTMALYVSLGISGGHLN 122

TbbAQP3 ELRLNYRDYMGELLGT FVLLFMGNGVATVIIDGKLGFLSITLWGI AVTMALYVSLGISGGHLN 122

*****:*****

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TbbAQP2 (+pentamidine) SAVTVGNVAVFGDFPWRKVPYIAAQMLGTFLGAACAYGVFADLLKAHGGGELIAFGEKGIWVFA 195

TbbAQP2 (+melarsoprol) SAVTVGNVAVFGDFPWRKVPYIAAQMLGTFLGAACAYGVFADLLKAHGGGELIAFGEKGIWVFA 195

TbgAQP2/3 (678-880) SAVTVGNVAVFGDFPWRKVPYIAAQMLGTFLGAACAYGVFADLLKAHGGGELIAFGEKGIWVFA 195

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TbbR25 PAVTVGNVAVFGDFPWRKVPYIAAQMLGAFLGAACAYGVFADLLKAHGGGELIAFGEKGIWVFA 187

TbbAQP3 PAVTVGNVAVFGDFPWRKVPYIAAQMLGAFLGAACAYGVFADLLKAHGGGELIAFGEKGTAGVFS 187

*****:*****

TbbAQP2 (+glycerol) MYPAEGNGIFYPIFAELISTAVLLLCVCGIFDPNNSPAKGYETVAIGALVFVMVNNFGLA 255

TbbAQP2 (+pentamidine) MYPAEGNGIFYPIFAELISTAVLLLCVCGIFDPNNSPAKGYETVAIGALVFVMVNNFGLA 255

TbbAQP2 (+melarsoprol) MYPAEGNGIFYPIFAELISTAVLLLCVCGIFDPNNSPAKGYETVAIGALVFVMVNNFGLA 255

TbgAQP2/3 (678-880) MYPAEGNGIFYPIFAELISTAVLLLCVCGIFDPNNSPAKGHEPLAVGALVFAIGNNIGYS 255

TbbP1000 TYPRDSNGLFSCIFGEFICTAMLLFCVCGIFDPNNSPAKGHEPLAVGALVFAIGNNIGYS 255

Tbb427MR TYPRDSNGLFSCIFGEFICTAMLLFCVCGIFDPNNSPAKGHEPLAVGALVFAIGNNIGYS 255

TbbR25 MYPAEGNGIFYPIFAELISTAVLLLCVCGIFDPNNSPAKGYETVAIGALVFVMVNNFGLA 247

TbbAQP3 TYPRDSNGLFSCIFGEFICTAMLLFCVCGIFDPNNSPAKGHEPLAVGALVFAIGNNIGYS 247

** :.*:*****:*****

TbbAQP2 (+glycerol) SPLAMNPSLDFGPRVFGAILLGGEVFSHANYFVWVPLVVPFFGAILGLFLYKYFLPH 312

TbbAQP2 (+pentamidine) SPLAMNPSLDFGPRVFGAILLGGEVFSHANYFVWVPLVVPFFGAILGLFLYKYFLPH 312

TbbAQP2 (+melarsoprol) SPLAMNPSLDFGPRVFGAILLGGEVFSHANYFVWVPLVVPFFGAILGLFLYKYFLPH 312

TbgAQP2/3 (678-880) TG~~Y~~AINPARDFGPRVFSFLYGGEVFSHANYFVWVPLVVPFFGAILGLFLYKYFLPH 312

TbbP1000 TG~~Y~~AINPARDFGPRVFSFLYGGEVFSHANYFVWVPLVVPFFGAILGLFLYKYFLPH 312

Tbb427MR TG~~Y~~AINPARDFGPRVFSFLYGGEVFSHANYFVWVPLVPLPFGGIFGLFLYKYFVPH 312

TbbR25 SPLAMNPSLDFGPRVFGAILLGGEVFSHANYFVWVPLVVPFFGAILGLFLYKYFLPH 304

TbbAQP3 TG~~Y~~AINPARDFGPRVFSFLYGGKVFVSHANYFVWVPLVPLPFGGIFGLFLYKYFVPH 304

: * :.*:*****:*****

Fig. S1

Alignment of aquaporin amino acid sequences.

Amino acid sequences of AQP2 and AQP3 from *Trypanosoma brucei brucei* (Tbb) are aligned with chimeric aquaporins from drug resistant strains (P1000, 427MR, R25) and a chimera from drug-resistant *T. b. gambiense* (TbgAQP2/3). Three identical sequences are shown of TbbAQP2 with residues making contact to either glycerol, pentamidine or melarsoprol highlighted (cyan, green or yellow, respectively). Red bars represent transmembrane regions. Residues highlighted in grey are conserved in all sequences. The conserved NSA/NPS motif in TbAQP2 and the analogous NPA/NPA motif in TbAQP3 are in purple font and the residues mutated for the MD simulation studies (I110W, L258Y and L264R) are in red. Sequence information was obtained from Munday et al.¹² except R25²⁸ and TbgAQP2/3 (678-800)¹⁵.

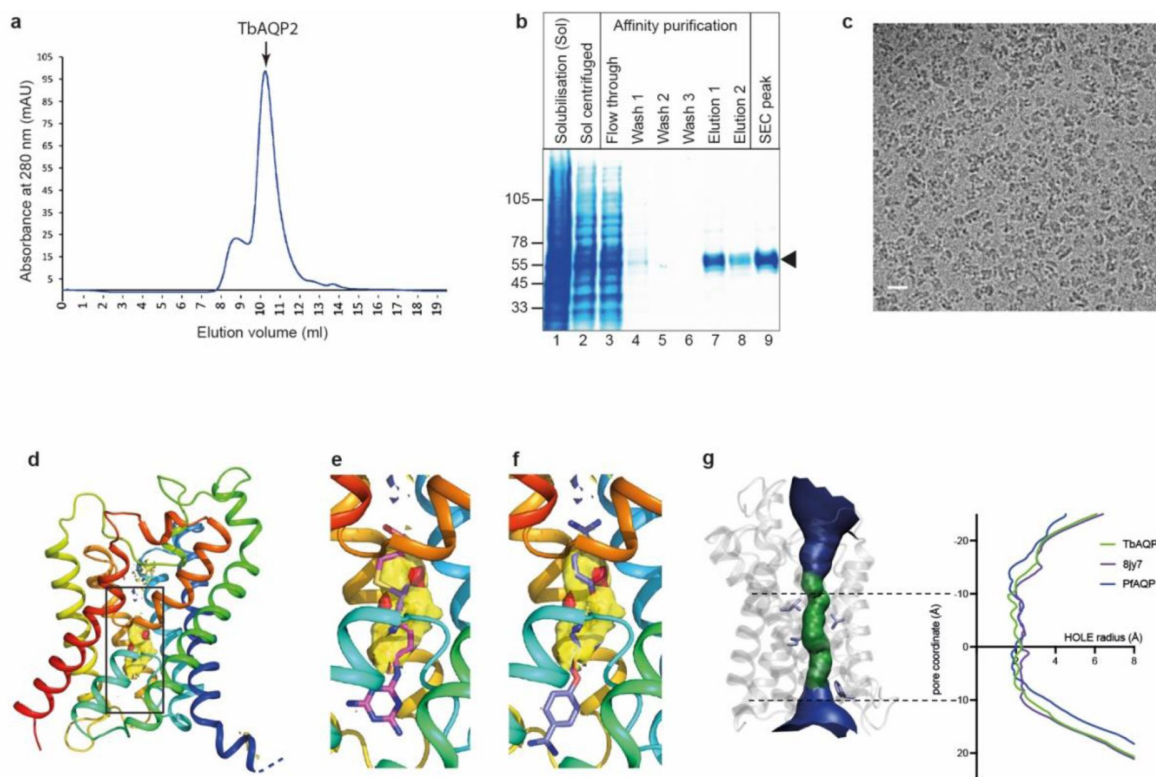


Fig. S2

Purification of TbAQP2 and identification of the drug binding site by hotspot analysis.

a, SEC trace of affinity purified TbAQP2. **b**, Coomassie-stained SDS-PAGE gel of fractions during detergent solubilisation (Sol), Ni^{2+} -affinity chromatography and from the SEC column. **c**, Cryo-EM electron micrograph showing the particle distribution of TbAQP2. **d**, A region of TbAQP2 was identified as being most likely to bind a ligand is shown (yellow surface) is mapped on to the structure of TbAQP2 determined here (rainbow colouration). **e**, as in **d**, but also containing pentamidine (cyan sticks). **f**, HOLE⁸⁷ trace for our resolved TbAQP2 structure, compared to a previously released TbAQP2 structure ('8yj7')³¹ and *Plasmodium falciparum* AQP²⁴.

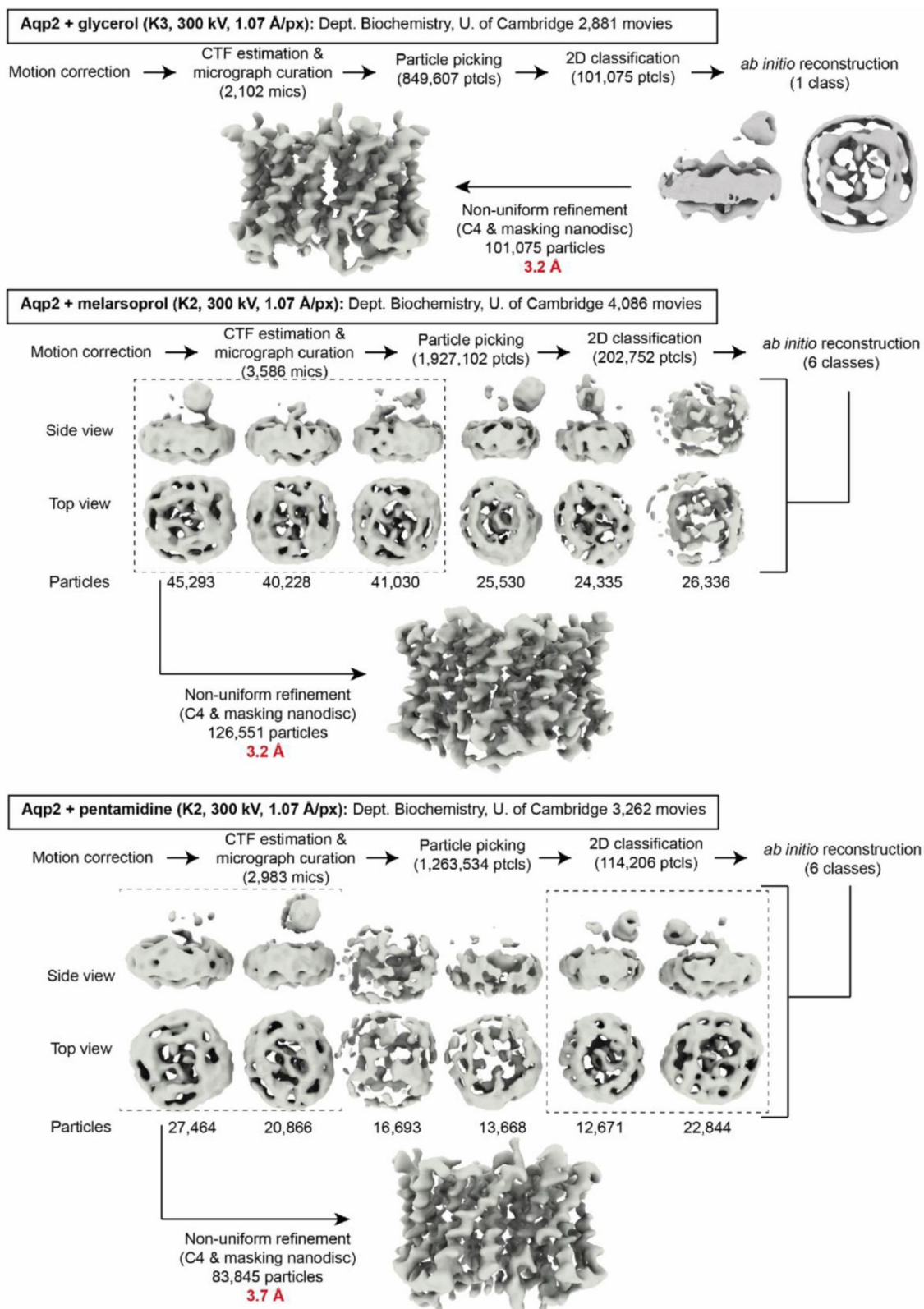


Fig. S3

Flow chart of the cryo-EM image processing and structure determination.

See the Methods section for the computer programmes used for each step and for further details.

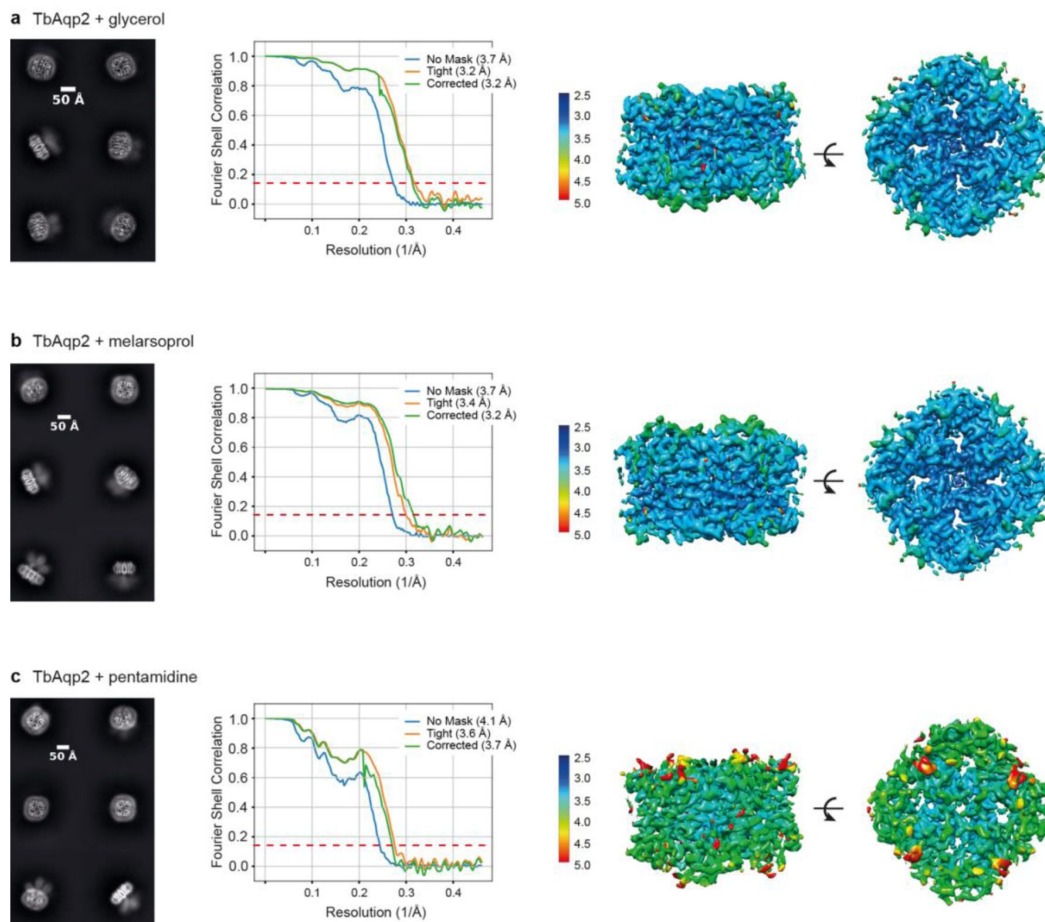
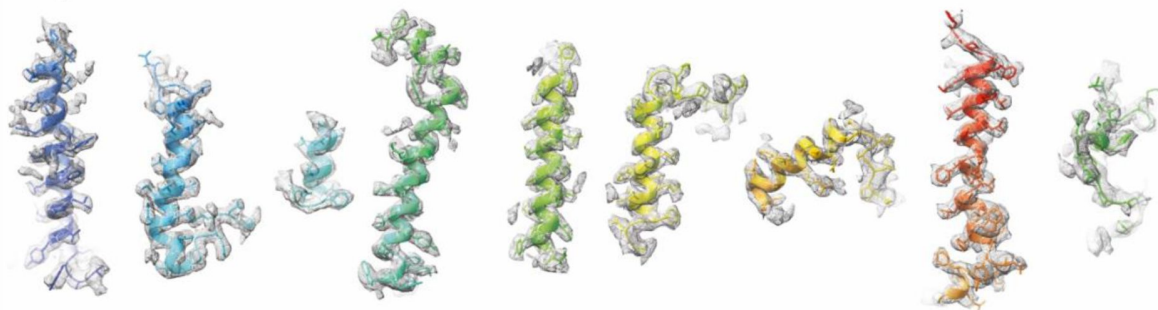


Fig. S4

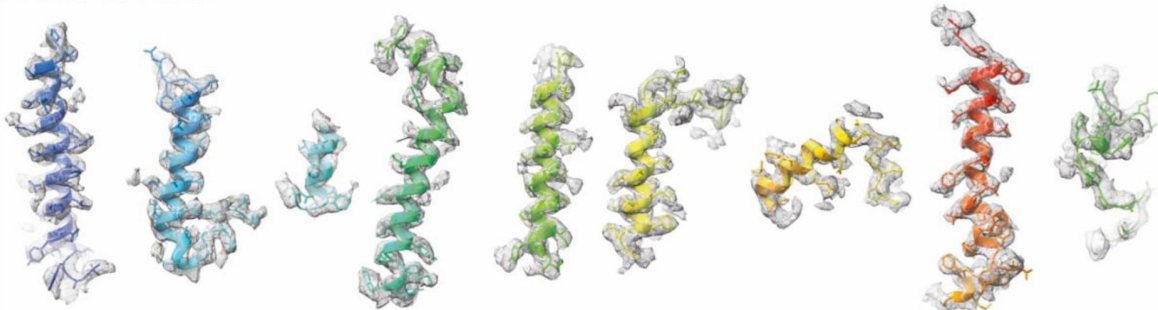
Local resolution maps of TbAQP2 structures.

a-c, 2D class averages for each of the structures (left panels), FSC curves of the reconstructions with estimates for resolution determined using an FSC of 0.143 as implemented in cryoSPARC (middle panels) and local resolution estimations as calculated by MonoRes (right panels).

Melarsoprol-bound TbAQP2



Pentamidine-bound TbAQP2



Glycerol-bound TbAQP2

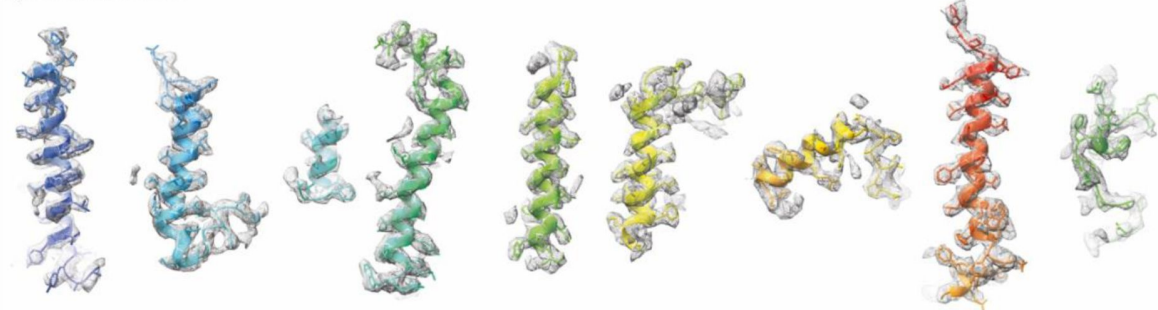
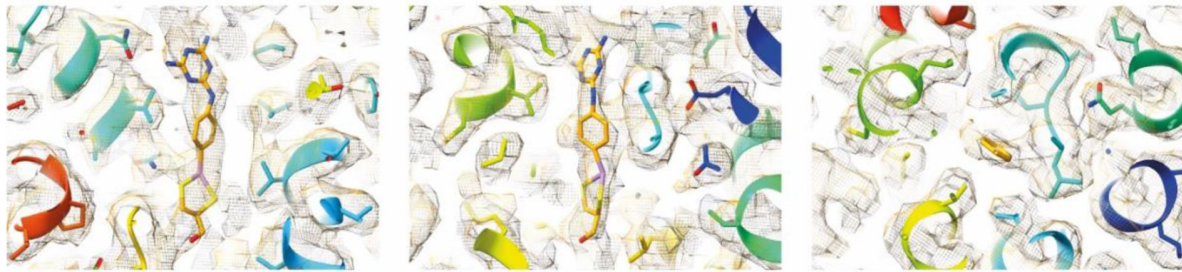


Fig. S5

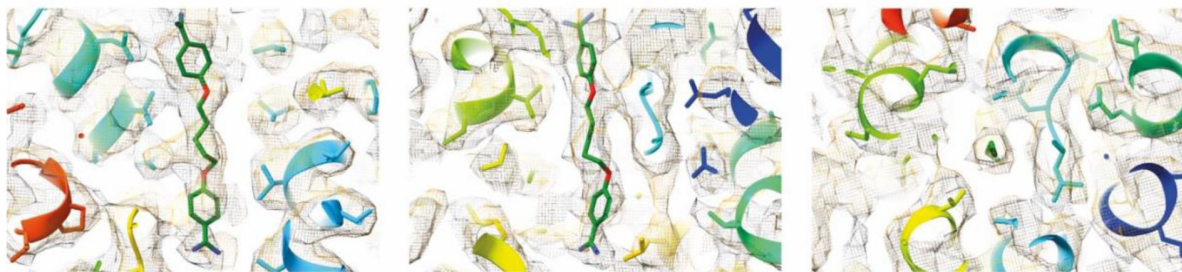
Density maps of secondary structure elements in the TbAQP2 structures.

Colouring is the same as in [Fig 1b](#).

Melarsoprol-bound TbAQP2



Pentamidine-bound TbAQP2



Glycerol-bound TbAQP2

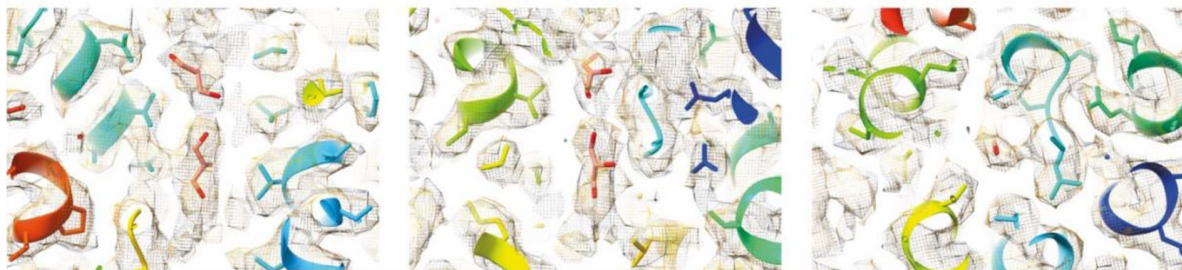


Fig. S6

Density half maps of ligands bound in the TbAQP2 structures.

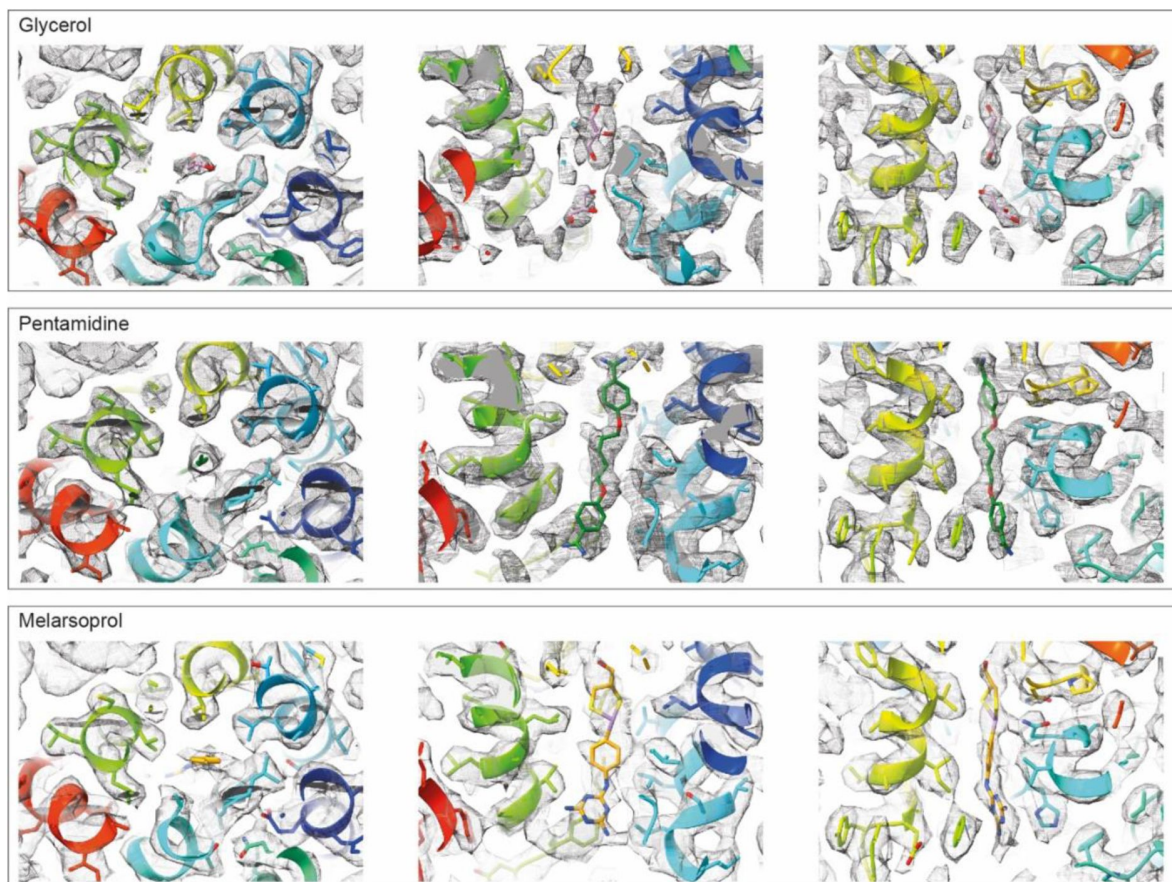
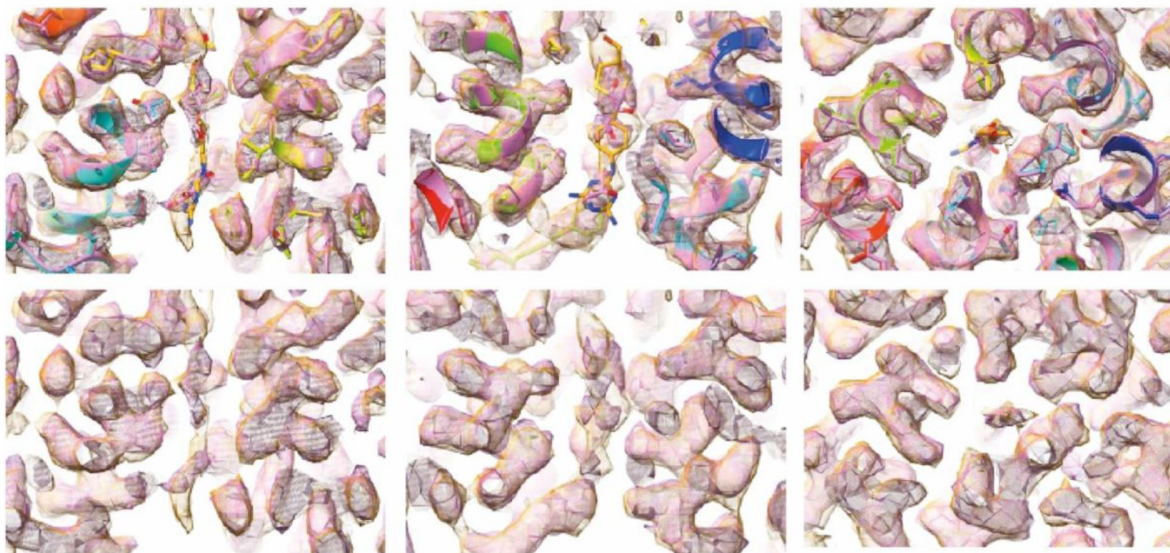


Fig. S7

Depictions of the side chain and substrate densities.

a Melarsoprol compared to glycerol



b Pentamidine compared to glycerol

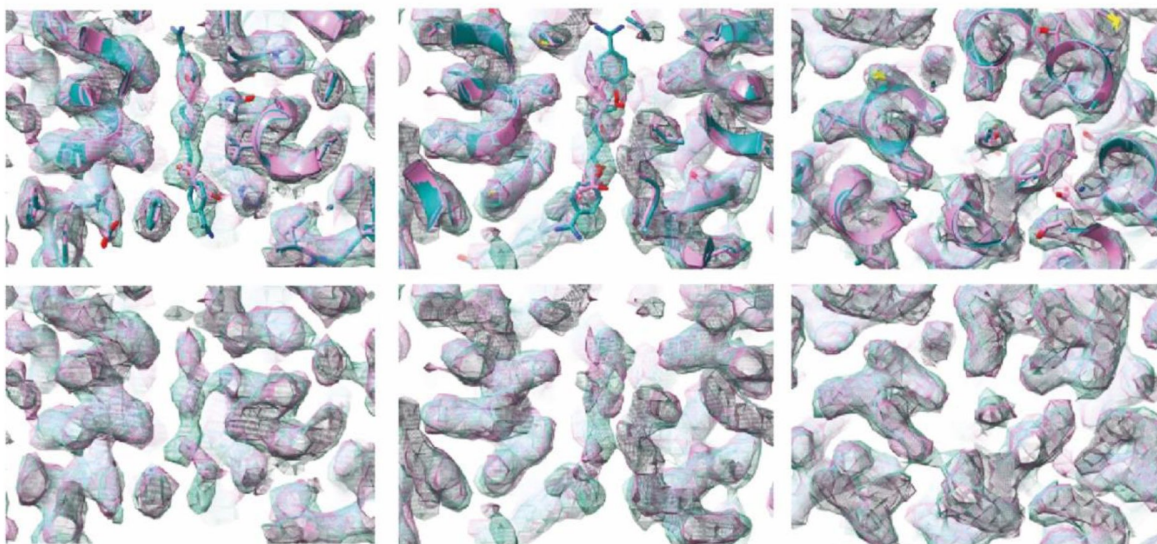


Fig. S8

Comparisons between substrate densities.

a, Comparison between melarsoprol and glycerol. **b**, comparison between pentamidine and glycerol.

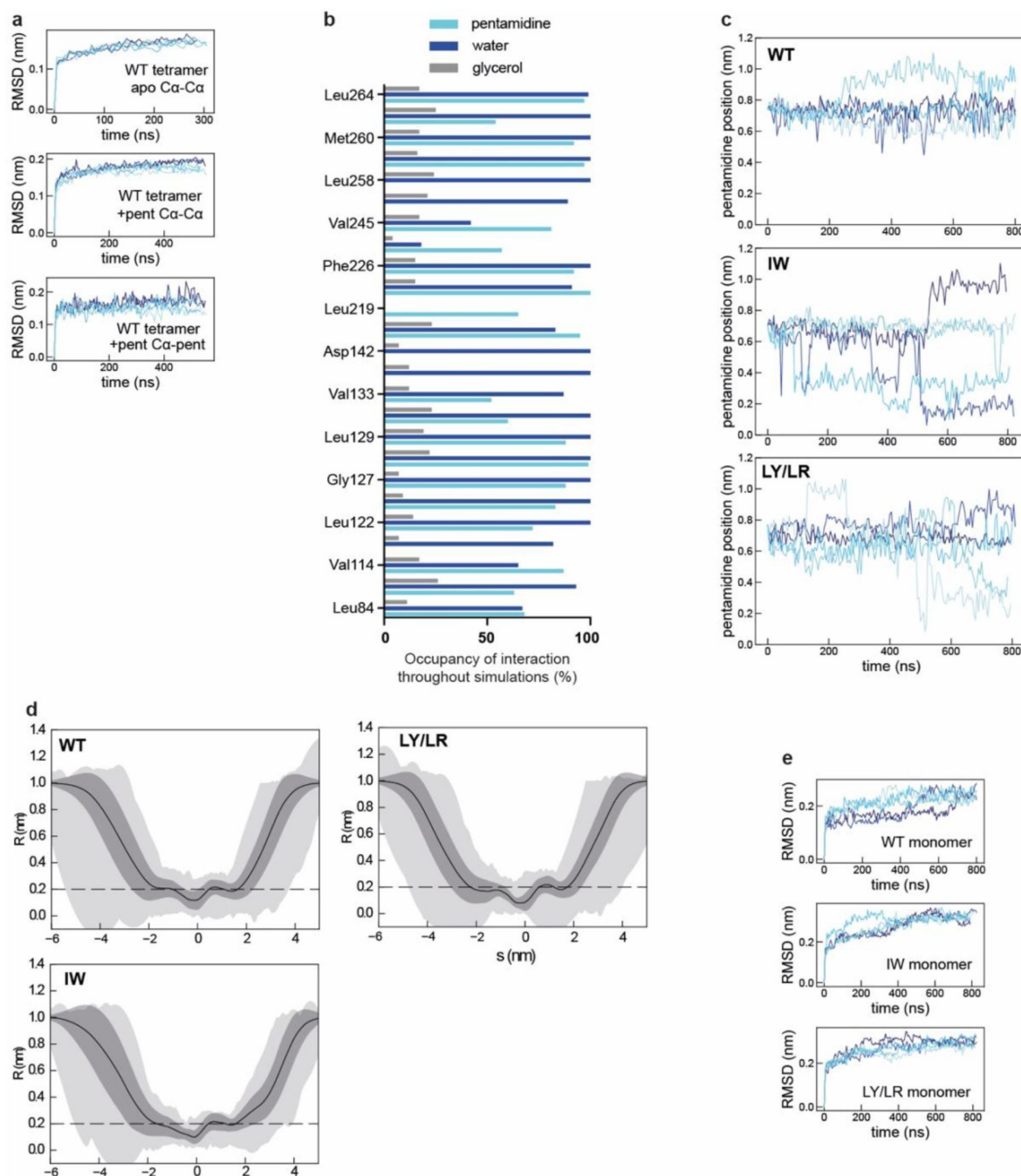


Fig. S9

Molecular dynamics simulations.

a, RMSDs for simulations of tetrameric WT TbAQP2. Trajectories were fitted on the Ca atoms and RMSDs were calculated for either Ca atoms for apo TbAQP2 (top), Ca atoms for pentamidine (pent) bound TbAQP2 (middle), or for the pentamidine molecule itself, i.e. in relation to the Ca of the channel (bottom). Five independent repeats are shown as blue traces. **b**, Percentage of time (occupancy) a given residue is within 3.9 Å of either pentamidine, water or glycerol during the MD simulations. **c**, Plotting the z-axis position of pentamidine when bound to monomeric TbAQP2 in a WT, IW, or LY/LR background. **d**, Pore radius profiles from simulations of apo monomeric WT, IW, or LY/LR TbAQP2. Analyses were run using the CHAP package⁸³ with traces showing the mean (black line), standard deviation (dark grey), and range (light grey) for all frames over 5 × 800 ns of simulation. **e**, As panel **a**, but for Ca atoms from monomeric TbAQP2 in a WT background, or with I110W (IW) or L258Y/L264R (LY/LR) mutations.

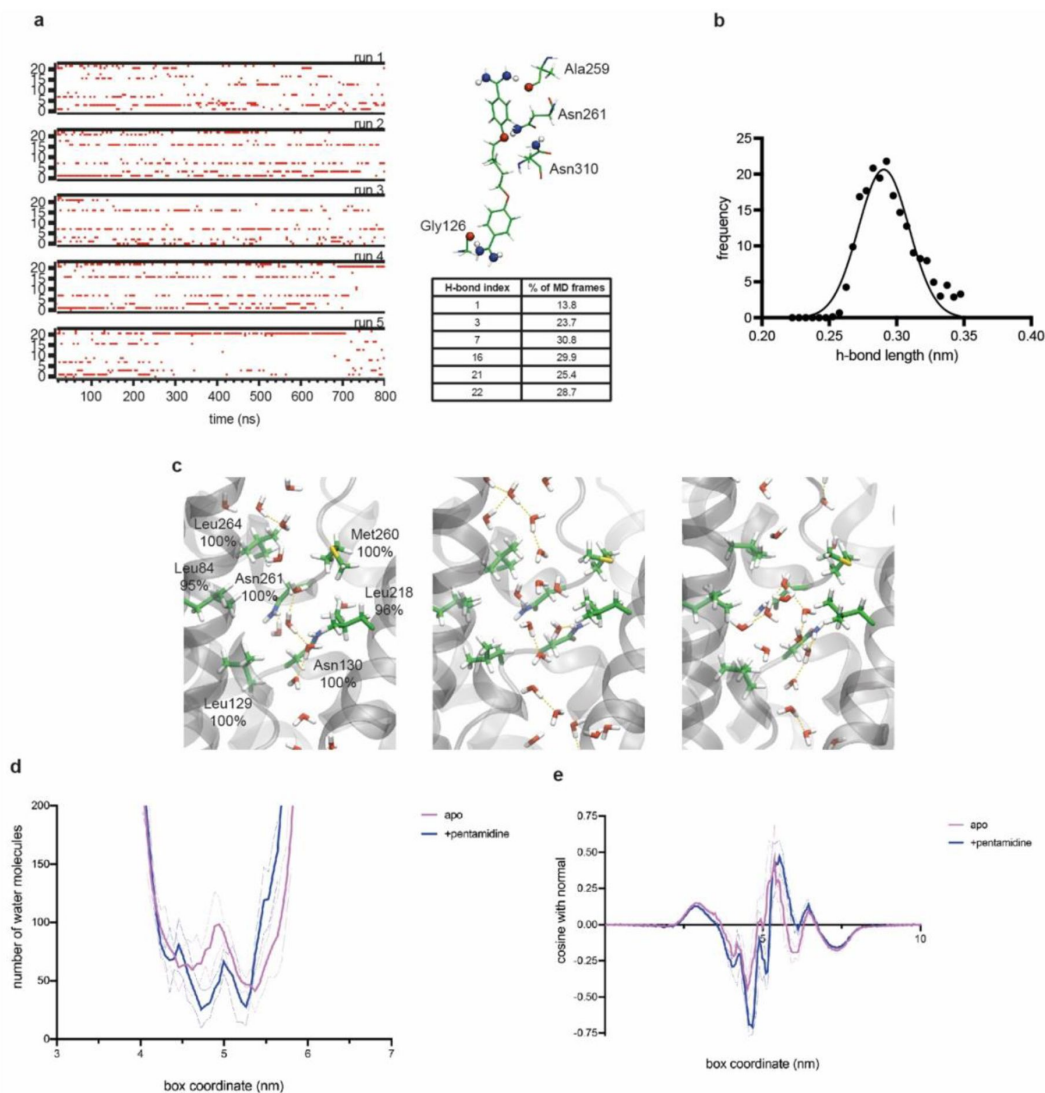


Fig. S10

Water in molecular dynamics simulations

a, hydrogen-bond analysis from *gmx hbond* between the pentamidine and channel. Across the 5 independent MD runs of the monomeric TbAQP2 MD simulations, 23 hydrogen-bonds are formed, but only 6 appear to be high occupancy. These are highlighted on the image along with their index number, and their quantification is shown below. **b**, hydrogen-bond distances from the bonds in panel **a** plotted as a frequency plot. The average bond length was 0.29 ± 0.02 nm, suggesting moderate strength. **c**, snapshots from MD showing water progression through the TbAQP2 channel in the absence of pentamidine. Three snapshots are shown. Residues that form especially high contact with water molecules (values given in brackets) are highlighted in each snapshot. Water-water hydrogen bonds, as computed using VMD, are shown in yellow dashed lines. **d**, number of water molecules along the TbAQP2 channel $\pm$ pentamidine, as computed using *gmx h2order* on the monomeric TbAQP2 MD simulations. **e**, orientation of the waters in each of the slices along with channel, as computed using *gmx h2order*.

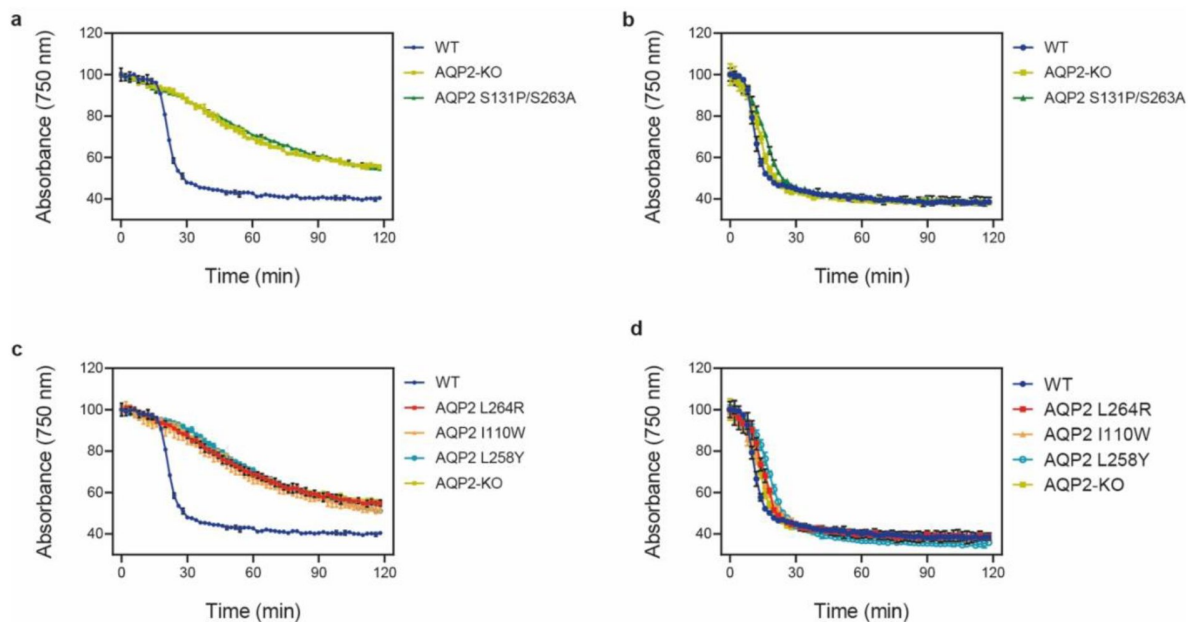


Fig. S11

Lysis assay of *T. brucei* cells.

a-d, Lysis assay with *T. brucei* wild-type and various AQP2 mutant cell lines treated with arsenical compounds: **a** and **c**, cymelarsan; **b** and **d**, phenylarsine oxide, a different arsenic-containing trypanocide that enters the parasite independently of TbAQP2^{8,10,12} and is thus used as a control for transporter-related arsenic resistance versus resistance to arsenic *per se*. The cells were placed in a cuvette and treated with either compound at $t = 10$ min. All points shown are the average of triplicate determinations and SD. When error bars are not visible, they fall within the symbol. The slow decline with cymelarsan over time in AQP2-KO and the mutant cell lines is attributable to residual uptake of the compound through the TbAT1/P2 transporter^{88,89}.

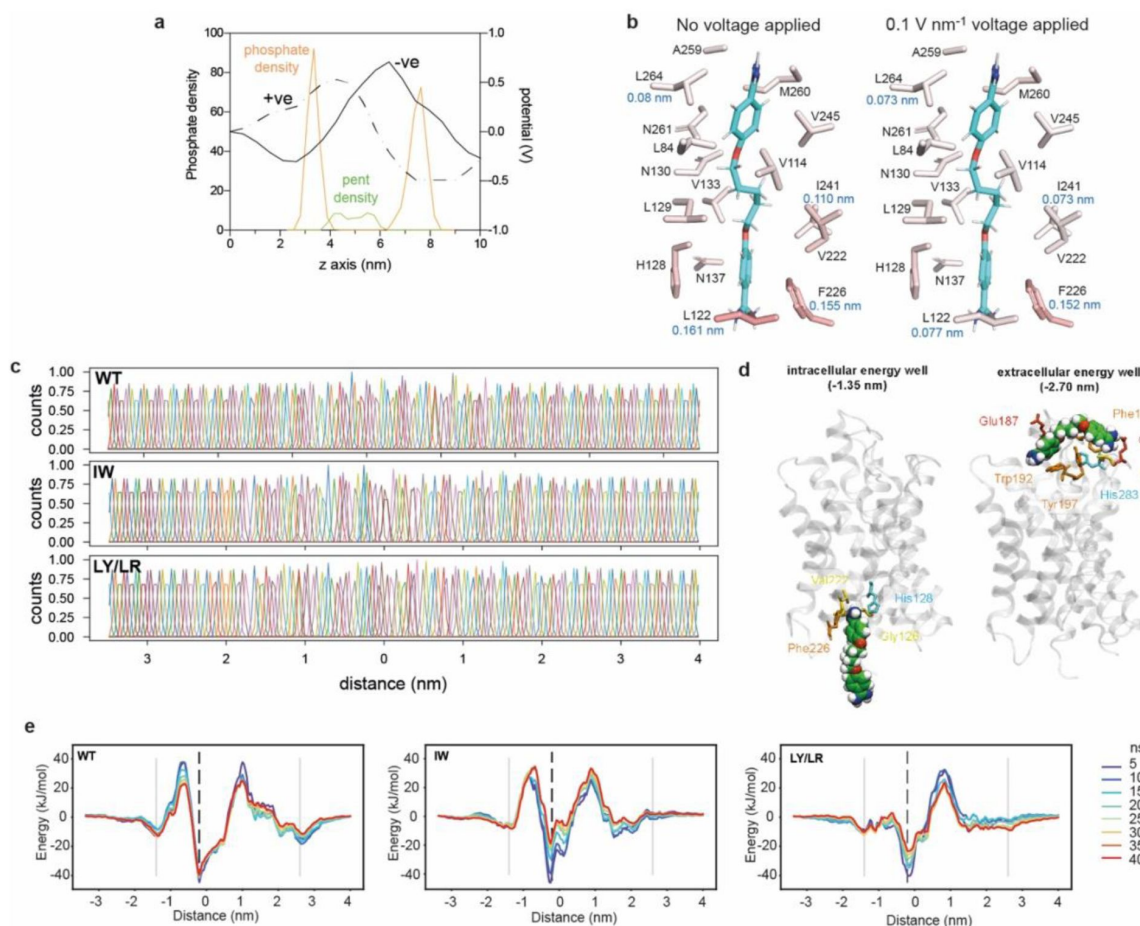


Fig. S12

Molecular dynamics simulations.

a, Molecular dynamics simulations. **a**, Plot of the membrane potential (V) as applied via an electric field, and calculated using the Gromacs tool *gmx potential*. Shown are potentials produced by an electric field applied in the forward (-ve; solid line) or backward (+ve; dashed line) direction. For context, the densities of lipid phosphate atoms and pent atoms are shown as orange and green traces, as measured using *gmx density*. **b**, RMSF calculations were run on monomeric TbAQP2 with either no membrane voltage or a 0.1 V nm^{-1} voltage applied (in the physiological direction). Shown are residues in contact with the pentamidine molecule, coloured by RMSF value. RMSF values are shown for residues Leu122, Phe226, Ile241, and Leu264. The data suggest the voltage has little impact on the flexibility or stability of the pore lining residues. **c**, Histograms for the umbrella sampling simulations used to generate the landscapes in **Fig. 4d**. The data show considerable overlap between windows. **d**, Snapshots taken from post umbrella sampling windows for WT TbAQP2 at -1.35 nm and +2.7 nm. **e**, Conversion plots for the PMF landscapes. Shown are landscapes calculated for increasing lengths of simulations (length in ns). After about 30 ns, the landscapes converged.

	TbAQP2 + glycerol (EMDB: EMD-16864) (PDB: 8OFZ)	TbAQP2 + melarsoprol (EMDB: EMD-16862) (PDB: 8OFX)	TbAQP2 + pentamidine (EMDB: EMD-16863) (PDB: 8OFY)
Data collection and processing			
Magnification	130,000	130,000	130,000
Voltage (kV)	300	300	300
Electron exposure (e ⁻ /Å ²)	25,2	57,6	57,6
Defocus range (μm)	[-0.6 , -4]	[-0.6 , -3.6]	[-0.8 , -4]
Pixel size (Å)	1.07	1.042*	1.042*
Symmetry imposed	C4	C4	C4
Initial particle images (no.)	849,607	1,927,102	1,263,534
Final particle images (no.)	101,075	126,551	83,845
Map resolution (Å)	3.2	3.2	3.7
FSC threshold	0.143	0.143	0.143
Map resolution range (Å)†	3.1 – 4.4	3.0 – 8.3	3.5 – 9.8
Refinement			
Initial model used	AlphaFold	AlphaFold	AlphaFold
Model composition in the ASU			
Non-hydrogen atoms	1842	1852	1855
Protein residues	244	244	244
Substrates	2	1	1
<i>B</i> factors (Å ²)			
Protein	159.7	140.2	170.4
Ligand	180.8	175.0	209.6
R.m.s. deviations			
Bond lengths (Å)	0.0074	0.0099	0.0060
Bond angles (°)	1.730	1.903	1.525
Validation			
MolProbity score	1.99	1.79	1.79
Clashscore	4.34	6.80	4.75
Poor rotamers (%)	5.52	1.66	2.21
Ramachandran plot			
Favored (%)	96.69	96.28	95.87
Allowed (%)	3.31	3.72	3.72
Disallowed (%)	0	0	0.41

Footnotes:

*Original pixel size in movies: 1.07 Å/px. Calibrated for modelling

†Local resolution range

Table S1

Cryo-EM data collection, refinement and validation statistics.

System	Box size	No. atoms	Setup	Simulation length
AQP2 WT tetramer	10x10x10 nm	ca. 95,000	Pentamidine	5 x ca. 300 ns
AQP2 WT tetramer	10x10x10 nm	ca. 95,000	Apo	5 x ca. 300 ns
AQP2 WT monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine	5 x 800 ns
AQP2 WT monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine and +ve electric field	3 x ca. 600 ns
AQP2 WT monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine and -ve electric field	3 x ca. 1400 ns
AQP2 WT monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine steered MD	1 x 5 ns pull
AQP2 WT monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine PMF	167 x 40 ns windows
AQP2 WT monomer	6.5x6.5x10 nm	ca. 45,000	Apo	5 x 800 ns
AQP2 _{L258Y_L264R} monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine	5 x 800 ns
AQP2 _{L258Y_L264R} monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine steered MD	1 x 5 ns pull
AQP2 _{L258Y_L264R} monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine PMF	167 x 40 ns windows
AQP2 _{L258Y_L264R} monomer	6.5x6.5x10 nm	ca. 45,000	Apo	5 x 800 ns
AQP2 _{I110W} monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine	5 x 800 ns
AQP2 _{I110W} monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine steered MD	1 x 5 ns pull
AQP2 _{I110W} monomer	6.5x6.5x10 nm	ca. 45,000	Pentamidine PMF	167 x 40 ns windows
AQP2 _{I110W} monomer	6.5x6.5x10 nm	ca. 45,000	Apo	5 x 800 ns

Table S2

Details of molecular dynamics simulations on TbAQP2.

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Additional information

Data availability

Structures have been deposited in the Protein Data Bank (PDB; <https://www.rcsb.org/>), and the associated cryo-EM data have been deposited in the Electron Microscopy Data Bank (EMDB; <https://www.ebi.ac.uk/pdbe/emdb/>) and the Electron Microscopy Public Image Archive (EMPIAR; <https://www.ebi.ac.uk/empair/>): SapA-TbAQP2-NP bound to glycerol (8OFZ, EMD-16864 and EMPIAR-11410); SapA-TbAQP2-NP bound to pentamidine (8OFY, EMD-16863 and EMPIAR-11412); SapA-TbAQP2-NP bound to melarsoprol (8OFX, EMD-16862 and EMPIAR-11411). There are no restrictions on data availability.

Author contributions

M.C. and S.N.W. conceived the project. T.S. and S.N.W. cloned, overexpressed, purified, and analysed protein samples. J.N.B., P.C.-H., D.Y.C. and S.N.W. prepared grids, screened and collected data. M.G. and R.M. processed the raw data. M.M. and P.A. built the models and performed refinements. A.G. performed further structural analyses. K.Y. and R.W. identified and fitted the substrates and performed refinements. J.M.C. advised on the processing and G.M. advised on the refinement process. V.S.C. performed hot spot analysis and T.L.B. advised. H.P.d.K. and M.C. advised the mutational studies. R.A.C. performed the molecular dynamics simulations, and R.A.C., P.S. and M.S. did the analysis of the MD simulation trajectories. M.A.U. performed cymelarsan uptake experiments. C.G.T. carried out structure analysis and S.N.W. managed the overall project. The manuscript was written by S.N.W., H.P.d.K., R.A.C. and C.G.T., and included contributions from all the authors.

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Reviewer #1 (Public review):

This study presents cryoEM-derived structures of the Trypanosome aquaporin AQP2, in complex with its natural ligand, glycerol, as well as two trypanocidal drugs, pentamidine and melarsoprol, which use AQP2 as an uptake route. The structures are high quality and the density for the drug molecules is convincing, showing a binding site in the centre of the AQP2 pore.

The authors then continue to study this system using molecular dynamics simulations. Their simulations indicate that the drugs can pass through the pore and identify a weak binding site in the centre of the pore, which corresponds with that identified through cryoEM analysis. They also simulate the effect of drug resistance mutations which suggests that the mutations reduce the affinity for drugs and therefore might reduce the likelihood that the drugs enter into the centre of the pore, reducing the likelihood that they progress through into the cell.

While the cryoEM and MD studies are well conducted, it is a shame that the drug transport hypothesis was not tested experimentally. For example, did they do cryoEM with AQP2 with drug resistance mutations and see if they could see the drugs in these maps? They might not bind, but another possibility is that the binding site shifts, as seen in Chen et al? Do they have an assay for measuring drug binding? I think that some experimental validation of the drug binding hypothesis would strengthen this paper. The authors describe in their response why these experiments are challenging.

<https://doi.org/10.7554/eLife.107460.2.sa3>

Reviewer #2 (Public review):

Summary:

The authors present 3.2-3.7 Å cryo-EM structures of *Trypanosoma brucei* aquaglyceroporin-2 (TbAQP2) bound to glycerol, pentamidine or melarsoprol and combine them with extensive all-atom MD simulations to explain drug recognition and resistance mutations. The work

provides a persuasive structural rationale for (i) why positively selected pore substitutions enable diamidine uptake, and (ii) how clinical resistance mutations weaken the high-affinity energy minimum that drives permeation. These insights are valuable for chemotherapeutic re-engineering of diamidines and aquaglyceroporin-mediated drug delivery.

My comments are on the MD part

Strengths:

The study

Integrates complementary cryo-EM, equilibrium and applied voltage MD simulations, and umbrella-sampling PMFs, yielding a coherent molecular-level picture of drug permeation.

Offers direct structural rationalisation of long-standing resistance mutations in trypanosomes, addressing an important medical problem.

Comments on revisions:

Most of the weaknesses have been resolved during the revision process.

<https://doi.org/10.7554/eLife.107460.2.sa2>

Reviewer #3 (Public review):

Summary:

Recent studies have established that trypanocidal drugs, including pentamidine and melarsoprol, enter the trypanosomes via the glyceroaquaporin AQP2 (TbAQP2). Interestingly, drug resistance in trypanosomes is, at least in part, caused by recombination with the neighbouring gene, AQP3, which is unable to permeate pentamidine or melarsoprol. The effect of the drugs on cells expressing chimeric proteins is significantly reduced. In addition, controversy exists regarding whether TbAQP2 permeates the drugs like an ion channel, or whether it serves as a receptor that triggers downstream processes upon drug binding. In this study the authors set out to achieve these objectives: 1) to understand the molecular interactions between TbAQP2 and glycerol, pentamidine, and melarsoprol, and 2) to determine the mechanism by which mutations that arise from recombination with TbAQP3 result in reduced drug permeation.

The cryo-EM structures provide details of glycerol and drug binding, and show that glycerol and the drugs occupy the same space within the pore. Finally, MD simulations and lysis assays are employed to determine how mutations in TbAQP2 result in reduced permeation of drugs by making entry and exit of the drug relatively more energy-expensive. Overall, the strength of evidence used to support the author's claims is solid.

Strengths:

The cryo-EM portion of the study is strong, and while the overall resolution of the structures is in the 3.5Å range, the local resolution within the core of the protein and the drug binding sites is considerably higher (~2.5Å).

I also appreciated the MD simulations on the TbAQP2 mutants and the mechanistic insights that resulted from this data.

Weaknesses:

(1) The authors do not provide any experimental validation the drug binding sites in TbAQP2 due to lacking resources. However, the claims have been softened in the revised paper.

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

This study presents cryoEM-derived structures of the Trypanosome aquaporin AQP2, in complex with its natural ligand, glycerol, as well as two trypanocidal drugs, pentamidine and melarsoprol, which use AQP2 as an uptake route. The structures are high quality, and the density for the drug molecules is convincing, showing a binding site in the centre of the AQP2 pore.

The authors then continue to study this system using molecular dynamics simulations. Their simulations indicate that the drugs can pass through the pore and identify a weak binding site in the centre of the pore, which corresponds with that identified through cryoEM analysis. They also simulate the effect of drug resistance mutations, which suggests that the mutations reduce the affinity for drugs and therefore might reduce the likelihood that the drugs enter into the centre of the pore, reducing the likelihood that they progress through into the cell.

While the cryoEM and MD studies are well conducted, it is a shame that the drug transport hypothesis was not tested experimentally. For example, did they do cryoEM with AQP2 with drug resistance mutations and see if they could see the drugs in these maps? They might not bind, but another possibility is that the binding site shifts, as seen in Chen et al.

TbAQP2 from the drug-resistant mutants does not transport either melarsoprol or pentamidine and there was thus no evidence to suggest that the mutant TbAQP2 channels could bind either drug. Moreover, there is not a single mutation that is characteristic for drug resistance in TbAQP2: references 12–15 show a plethora of chimeric AQP2/3 constructs in addition to various point mutations in laboratory strains and field isolates. In reference 17 we describe a substantial number of SNPs that reduced pentamidine and melarsoprol efficacy to levels that would constitute clinical resistance to acceptable dosage regimen. It thus appears that there are many and diverse mutations that are able to modify the protein sufficiently to induce resistance, and likely in multiple different ways, including the narrowing of the pore, changes to interacting amino acids, access to the pore etc. We therefore did not attempt to determine the structures of the mutant channels because we did not think that in most cases we would see any density for the drugs in the channel, and we would be unable to define ‘the’ resistance mechanism if we did in the case of one individual mutant TbAQP2. Our MD data suggests that pentamidine binding affinity is in the range of 50–300 μ M for the mutant TbAQP2s selected for that test (I110W and L258Y/L264R), i.e. >1000-fold higher than TbAQP2WT. Thus these structures will be exceedingly challenging to determine with pentamidine in the pore but, of course, until the experiment has been tried we will not know for sure.

Do they have an assay for measuring drug binding?

We tried many years ago to develop a 3 H-pentamidine binding assay to purified wild type TbAQP2 but we never got satisfactory results even though the binding should be in the doubledigit nanomolar range. This may be for any number of technical reasons and could also be partly because flexible di-benzamidines bind non-specifically to proteins at μ M concentrations giving rise to high background. Measuring binding to the mutants was not

tested given that they would be binding pentamidine in the μM range. If we were to pursue this further, then isothermal titration calorimetry (ITC) may be one way forward as this can measure μM affinity binding using unlabelled compounds, although it uses a lot of protein and background binding would need to be carefully assessed; see for example our work on measuring tetracycline binding to the tetracycline antiporter TetAB (<https://doi.org/10.1016/j.bbamem.2015.06.026>). Membrane proteins are also particularly tricky for this technique as the chemical activity of the protein solution must be identical to the chemical activity of the substrate solution which titrates in the molecule binding to the protein; this can be exceedingly problematic if any free detergent remains in the purified membrane protein. Another possibility may be fluorescence polarisation spectroscopy, although this would require fluorescently labelling the drugs which would very likely affect their affinity for TbAQP2 and how they interact with the wild type and mutant proteins – see the detailed SAR analysis in Alghamdi et al. 2020 (ref. 17). As you will appreciate, it would take considerable time and effort to set up an assay for measuring drug binding to mutants and is beyond the current scope of the current work.

I think that some experimental validation of the drug binding hypothesis would strengthen this paper. Without this, I would recommend the authors to soften the statement of their hypothesis (i.e, lines 65-68) as this has not been experimentally validated.

We agree with the referee that direct binding of drugs to the mutants would be very nice to have, but we have neither the time nor resources to do this. We have therefore softened the statement on lines 65-68 to read ‘Drug-resistant TbAQP2 mutants are still predicted to bind pentamidine, but the much weaker binding in the centre of the channel observed in the MD simulations would be insufficient to compensate for the high energy processes of ingress and egress, hence impairing transport at pharmacologically relevant concentrations.’

Reviewer #2 (Public review):

Summary:

The authors present 3.2-3.7 Å cryo-EM structures of Trypanosoma brucei aquaglyceroporin-2 (TbAQP2) bound to glycerol, pentamidine, or melarsoprol and combine them with extensive allatom MD simulations to explain drug recognition and resistance mutations. The work provides a persuasive structural rationale for (i) why positively selected pore substitutions enable diamidine uptake, and (ii) how clinical resistance mutations weaken the high-affinity energy minimum that drives permeation. These insights are valuable for chemotherapeutic re-engineering of diamidines and aquaglyceroporin-mediated drug delivery.

My comments are on the MD part.

Strengths:

The study

(1) Integrates complementary cryo-EM, equilibrium, applied voltage MD simulations, and umbrella-sampling PMFs, yielding a coherent molecular-level picture of drug permeation.

(2) Offers direct structural rationalisation of long-standing resistance mutations in trypanosomes, addressing an important medical problem.

Weaknesses:

Unphysiological membrane potential. A field of 0.1 V nm^{-1} ($\sim 1 \text{ V}$ across the bilayer) was applied to accelerate translocation. From the traces (Figure 1c), it can be seen that the translocation occurred really quickly through the channel, suggesting that the field might have introduced some large changes in the protein. The authors state that they checked visually for this, but some additional analysis, especially of the residues next to the drug, would be welcome.

This is a good point from the referee, and we thank them for raising it. It is common to use membrane potentials in simulations that are higher than the physiological value, although these are typically lower than used here. The reason we used the higher value was to speed sampling and it still took 1,400 ns for transport in the physiologically correct direction, and even then, only in 1/3 repeats. Hence this choice of voltage was probably necessary to see the effect. The exceedingly slow rate of pentamidine permeation seen in the MD simulation was consistent with the experimental observations, as discussed in Alghamdi et al (2020) [ref. 17] where we estimated that TbAQP2-mediated pentamidine uptake in *T. brucei* bloodstream forms proceeds at just 9.5×10^5 molecules/cell/h; the number of functional TbAQP2 units in the plasma membrane is not known but their location is limited to the small flagellar pocket (Quintana et al. PLoS Negl Trop Dis 14, e0008458 (2020)).

The referee is correct that it is important to make sure that the applied voltage is not causing issues for the protein, especially for residues in contact with the drug. We have carried out RMSF analysis to better test this. The data show that comparing our simulations with the voltage applied to the monomeric MD simulations + PNTM with no voltage reveals little difference in the dynamics of the drug-contacting residues.

We have added these new data as Supplementary Fig12b with a new legend (lines1134-1138)

‘b, RMSF calculations were run on monomeric TbAQP2 with either no membrane voltage or a 0.1 V nm^{-1} voltage applied (in the physiological direction). Shown are residues in contact with the pentamidine molecule, coloured by RMSF value. RMSF values are shown for residues Leu122, Phe226, Ile241, and Leu264. The data suggest the voltage has little impact on the flexibility or stability of the pore lining residues.’

We have also added the following text to the manuscript (lines 524-530):

‘Membrane potential simulations were run using the computational electrophysiology protocol. An electric field of 0.1 V/nm was applied in the z-axis dimension only, to create a membrane potential of about 1 V (see Fig. S10a). Note that this is higher than the physiological value of $87.1 \pm 2.1 \text{ mV}$ at pH 7.3 in bloodstream *T. brucei*, and was chosen to improve the sampling efficiency of the simulations. The protein and lipid molecules were visually confirmed to be unaffected by this voltage, which we quantify using RMSF analysis on pentamidine-contacting residues (Fig. S12b).’

Based on applied voltage simulations, the authors argue that the membrane potential would help get the drug into the cell, and that a high value of the potential was applied merely to speed up the simulation. At the same time, the barrier for translocation from PMF calculations is $\sim 40 \text{ kJ/mol}$ for WT. Is the physiological membrane voltage enough to overcome this barrier in a realistic time? In this context, I do not see how much value the applied voltage simulations have, as one can estimate the work needed to translocate the substrate on PMF profiles alone. The authors might want to tone down their conclusions about the role of membrane voltage in the drug translocation.

We agree that the PMF barriers are considerable, however we highlight that other studies have seen similar landscapes, e.g. PMID 38734677 which saw a barrier of ca. 10-15 kcal/mol

(ca. 4060 kJ/mol) for PNTM transversing the channel. This was reduced by ca. 4 kcal/mol when a 0.4 V nm^{-1} membrane potential was applied, so we expect a similar effect to be seen here.

We have updated the Results to more clearly highlight this point and added the following text (lines 274-275):

We note that previous studies using these approaches saw energy barriers of a similar size, and that these are reduced in the presence of a membrane voltage[17,31].’

Pentamidine charge state and protonation. The ligand was modeled as +2, yet pKa values might change with the micro-environment. Some justification of this choice would be welcome.

Pentamidine contains two diamidine groups and each are expected to have a pKa above 10 in solution (PMID: 20368397), suggesting that the molecule will carry a +2 charge. Using the +2 charge is also in line with previous MD studies (PMID: 32762841). We have added the following text to the Methods (lines 506-509):

‘The pentamidine molecule used existing parameters available in the CHARMM36 database under the name PNTM with a charge state of +2 to reflect the predicted pKas of >10 for these groups [73] and in line with previous MD studies[17].’

We note that accounting for the impact of the microenvironment is an excellent point – future studies might employ constant pH calculations to address this.

The authors state that this RMSD is small for the substrate and show plots in Figure S7a, with the bottom plot being presumably done for the substrate (the legends are misleading, though), levelling off at ~0.15 nm RMSD. However, in Figure S7a, we see one trace (light blue) deviating from the initial position by more than 0.2 nm - that would surely result in an RMSD larger than 0.15, but this is somewhat not reflected in the RMSD plots.

The bottom plot of Fig. S9a (previously Fig. S7a) is indeed the RMSD of the drug (in relation to the protein). We have clarified the legend with the following text (lines 1037-1038): ‘... or for the pentamidine molecule itself, i.e. in relation to the Ca of the channel (bottom).’

With regards the second comment, we assume the referee is referring to the light blue trace from Fig S9c. These data are actually for the monomeric channel rather than the tetramer. We apologise for not making this clearer in the legend. We have added the word ‘monomeric’ (line 1041).

Reviewer #3 (Public review):

Summary:

Recent studies have established that trypanocidal drugs, including pentamidine and melarsoprol, enter the trypanosomes via the glycerol aquaporin AQP2 (TbAQP2). Interestingly, drug resistance in trypanosomes is, at least in part, caused by recombination with the neighbouring gene, AQP3, which is unable to permeate pentamidine or melarsoprol. The effect of the drugs on cells expressing chimeric proteins is significantly reduced. In addition, controversy exists regarding whether TbAQP2 permeates drugs like an ion channel, or whether it serves as a receptor that triggers downstream processes upon drug binding. In this study the authors set out to achieve three objectives:

(1) to determine if TbAQP2 acts as a channel or a receptor,

We should clarify here that this was not an objective of the current manuscript as the transport activity has already been extensively characterised in the literature, as described in the introduction.

(2) to understand the molecular interactions between TbAQP2 and glycerol, pentamidine, and melarsoprol, and

(3) to determine the mechanism by which mutations that arise from recombination with TbAQP3 result in reduced drug permeation.

Indeed, all three objectives are achieved in this paper. Using MD simulations and cryo-EM, the authors determine that TbAQP2 likely permeates drugs like an ion channel. The cryo-EM structures provide details of glycerol and drug binding, and show that glycerol and the drugs occupy the same space within the pore. Finally, MD simulations and lysis assays are employed to determine how mutations in TbAQP2 result in reduced permeation of drugs by making entry and exit of the drug relatively more energy-expensive. Overall, the strength of evidence used to support the author's claims is solid.

Strengths:

The cryo-EM portion of the study is strong, and while the overall resolution of the structures is in the 3.5Å range, the local resolution within the core of the protein and the drug binding sites is considerably higher (~2.5Å).

I also appreciated the MD simulations on the TbAQP2 mutants and the mechanistic insights that resulted from this data.

Weaknesses:

(1) The authors do not provide any empirical validation of the drug binding sites in TbAQP2. While the discussion mentions that the binding site should not be thought of as a classical fixed site, the MD simulations show that there's an energetically preferred slot (i.e., high occupancy interactions) within the pore for the drugs. For example, mutagenesis and a lysis assay could provide us with some idea of the contribution/importance of the various residues identified in the structures to drug permeation. This data would also likely be very valuable in learning about selectivity for drugs in different AQP proteins.

On a philosophical level, we disagree with the requirement for ‘validation’ of a structure by mutagenesis. It is unclear what such mutagenesis would tell us beyond what was already shown experimentally through ³H-pentamidine transport, drug sensitivity and lysis assays i.e. a given mutation will impact permeation to a certain extent. But on the structural level, what does mutagenesis tell us? If a bulky aromatic residue that makes many van der Waals interactions with the substrate is changed to an alanine residue and transport is reduced, what does this mean? It would confirm that the phenylalanine residue is very likely indeed making van der Waals contacts to the substrate, but we knew that already from the WT structure. And if it doesn't have any effect? Well, it could mean that the van der Waals interactions with that particular residue are not that important or it could be that the substrate has changed its positions slightly in the channel and the new pose has similar energy of interactions to that observed in the wild type channel. Regardless of the result, any data from mutagenesis would be open to interpretation and therefore would not impact on the conclusions drawn in this manuscript. We might not learn anything new unless all residues interacting with the substrate are mutated, the structure of each mutant was determined and MD simulations were performed for all, which is beyond the scope of this work. Even then, the value for understanding clinical drug resistance would be limited, as this phenomenon has been linked to various chimeric rearrangements with adjacent TbAQP3

(references 12–15), each with a structure distinct from TbAQP2 with a single SNP. We also note that the recent paper by Chen et al. did not include any mutagenesis of the drug binding sites in TbAQP2 in their analysis of TbAQP2, presumably for similar reasons as discussed above.

(2) Given the importance of AQP3 in the shaping of AQP2-mediated drug resistance, I think a figure showing a comparison between the two protein structures/AlphaFold structures would be beneficial and appropriate

We agree that the comparison is of considerable interest and would contribute further to our understanding of the unique permeation capacities of TbAQP2. As such, we followed the reviewer's suggestion and made an AlphaFold model of TbAQP3 and compared it to our structures of TbAQP2. The RMSD is 0.6 Å to the pentamidine-bound TbAQP2, suggesting that the fold of TbAQP3 has been predicted well, although the side chain rotamers cannot be assessed for their accuracy. Previous work has defined the selectivity filter of TbAQP3 to be formed by W102, R256, Y250. The superposition of the TbAQP3 model and the TbAQP2 pentamidine-bound structure shows that one of the amine groups is level with R256 and that there is a clash with Y250 and the backbone carbonyl of Y250, which deviates in position from the backbone of TbAQP2 in this region. There is also a clash with Ile252.

Although these observations are indeed interesting, on their own they are highly preliminary and extensive further work would be necessary to draw any convincing conclusions regarding these residues in preventing uptake of pentamidine and melarsoprol. The TbAQP3 AlphaFold model would need to be verified by MD simulations and then we would want to look at how pentamidine would interact with the channel under different experimental conditions like we have done with TbAQP2. We would then want to mutate to Ala each of the residues singly and in combination and assess them in uptake assays to verify data from the MD simulations. This is a whole new study and, given the uncertainties surrounding the observations of just superimposing TbAQP2 structure and the TbAQP3 model, we feel that, regrettably, this is just too speculative to add to our manuscript.

(3) A few additional figures showing cryo-EM density, from both full maps and half maps, would help validate the data.

Two new Supplementary Figures have been made, on showing the densities for each of the secondary structure elements (the new Figure S5) and one for the half maps showing the ligands (the new Figure S6). All the remaining supplementary figures have been renamed accordingly.

(4) Finally, this paper might benefit from including more comparisons with and analysis of data published in Chen et al (doi.org/10.1038/s41467-024-48445-4), which focus on similar objectives. Looking at all the data in aggregate might reveal insights that are not obvious from either paper on their own. For example, melarsoprol binds differently in structures reported in the two respective papers, and this may tell us something about the energy of drug-protein interactions within the pore.

We already made the comparisons that we felt were most pertinent and included a figure (Fig. 5) to show the difference in orientation of melarsoprol in the two structures. We do not feel that any additional comparison is sufficiently interesting to be included. As we point out, the structures are virtually identical (RMSD 0.6 Å) and therefore there are no further mechanistic insights we would like to make beyond the thorough discussion in the Chen et al paper.

Reviewer #1 (Recommendations for the authors):

(1) Line 65 - I don't think that the authors have tested binding experimentally, and so rather than 'still bind', I think that 'are still predicted to bind' is more appropriate.

Changed as suggested

(2) Line 69 - remove 'and'

Changed as suggested

(3) Line 111 - clarify that it is the protein chain which is 'identical'. Ligands not.

Changed to read “The cryo-EM structures of TbAQP2 (excluding the drugs/substrates) were virtually identical...”

(4) Line 186 - make the heading of this section more descriptive of the conclusion than the technique?

We have changed the heading to read: ‘Molecular dynamics simulations show impaired pentamidine transport in mutants’

Reviewer #2 (Recommendations for the authors):

(1) Methods - a rate of 1 nm per ns is mentioned for pulling simulations, is that right?

Yes, for the generation of the initial frames for the umbrella sampling a pull rate of 1 nm/ns was used in either an upwards or downwards z-dimension

(2) Figure S9 and S10 have their captions swapped.

The captions have been swapped to their proper positions.

(3) Methods state "40 ns per window" yet also that "the first 50 ns of each window was discarded as equilibration".

Well spotted - this line should have read “the first 5 ns of each window was discarded as equilibration”. This has been corrected (line 541).

Reviewer #3 (Recommendations for the authors):

(1) Abstract, line 68-70: incomplete sentence.

The sentence has been re-written: “The structures of drug-bound TbAQP2 represent a novel paradigm for drug-transporter interactions and are a new mechanism for targeting drugs in pathogens and human cells.

(2) Line 312-313: The paper you mention here came out in May 2024 - a year ago. I appreciate that they reported similar structural data, but for the benefit of the readers and the field, I would recommend a more thorough account of the points by which the two pieces of work differ. Is there some knowledge that can be gleaned by looking at all the data in the two papers together? For example, you report a glycerol-bound structure while the other group provides an apo one. Are there any mechanistic insights that can be gained from a comparison?

We already made the comparisons that we felt were most pertinent and included a figure (Fig. 5) to show the difference in orientation of melarsoprol in the two structures. We do not feel that any additional comparison is sufficiently interesting to be included. As we point out, the structures are virtually identical (RMSD 0.6 Å) and therefore there are no further mechanistic insights we would like to make beyond the thorough discussion in the Chen et al paper.

(3) Similarly, you can highlight the findings from your MD simulations on the TbAQP2 drug resistance mutants, which are unique to your study. How can this data help with solving the drug resistance problem?

New drugs will need to be developed that can be transported by the mutant chimera AQP2s and the models from the MD simulations will provide a starting point for molecular docking studies. Further work will then be required in transport assays to optimise transport rather than merely binding. However, the fact that drug resistance can also arise through deletion of the AQP2 gene highlights the need for developing new drugs that target other proteins.

(4) A glaring question that one has as a reader is why you have not attempted to solve the structures of the drug resistance mutants, either in complex with the two compounds or in their apo/glycerol-bound form? To be clear, I am not requesting this data, but it might be a good idea to bring this up in the discussion.

TbAQP2 containing the drug-resistant mutants does not transport either melarsoprol or pentamidine (Munday et al., 2014; Alghamdi et al., 2020); there was thus no evidence to suggest that the mutant TbAQP2 channels could bind either drug. We therefore did not attempt to determine the structures of the mutant channels because we did not think that we would see any density for the drugs in the channel. Our MD data suggests that pentamidine binding affinity is in the range of 50-300 μM for the mutant TbAQP2, supporting the view that getting these structures would be highly challenging, but of course until the experiment is tried we will not know for sure.

We also do not think we would learn anything new about doing structures of the drug-free structures of the transport-negative mutants of TbAQP2. The MD simulations have given novel insights into why the drugs are not transported and we would rather expand effort in this direction and look at other mutants rather than expend further effort in determining new structures.

(5) Line 152-156: Is there a molecular explanation for why the TbAQP2 has 2 glycerol molecules captured in the selectivity filter while the PfAQP2 and the human AQP7 and AQP10 have 3?

The presence of glycerol molecules represents local energy minima for binding, which will depend on the local disposition of appropriate hydrogen bonding atoms and hydrophobic regions, in conjunction with the narrowness of the channel to effectively bind glycerol from all sides. It is noticeable that the extracellular region of the channel is wider in TbAQP2 than in AQP7 and AQP10, so this may be one reason why additional ordered glycerol molecules are absent, and only two are observed. Note also that the other structures were determined by X-ray crystallography, and the environment of the crystal lattice may have significantly decreased the rate of diffusion of glycerol, increasing the likelihood of observing their electron densities.

(6) I would also think about including the 8JY7 (TbAQP2 apo) structure in your analysis.

We included 8JY7 in our original analyses, but the results were identical to 8JY6 and 8JY8 in terms of the protein structure, and, in the absence of any modelled substrates in 8JY7 (the interesting part for our manuscript), we therefore have not included the comparison.

(7) I also think, given the importance of AQP3 in this context, it would be really useful to have a comparison with the AQP3 AlphaFold structure in order to examine why it does not permeate drugs.

We made an AlphaFold model of TbAQP3 and compared it to our structures of TbAQP2. The RMSD is 0.6 Å to the pentamidine-bound TbAQP2, suggesting that the fold of TbAQP3 has been predicted well, although the side chain rotamers cannot be assessed for their accuracy. Previous work has defined the selectivity filter of TbAQP3 to be formed by W102, R256, Y250. The superposition of the TbAQP3 model and the TbAQP2 pentamidine-bound structure shows that one of the amine groups is level with R256 and that there is a clash with Y250 and the backbone carbonyl of Y250, which deviates in position from the backbone of TbAQP2 in this region. There is also a clash with Ile252.

Although these observations are interesting, on their own they are preliminary in the extreme and extensive further work will be necessary to draw any convincing conclusions regarding these residues in preventing uptake of pentamidine and melarsoprol. The TbAQP3 AlphaFold model would need to be verified by MD simulations and then we would want to look at how pentamidine would interact with the channel under different experimental conditions like we have done with TbAQP2. We would then want to mutate to Ala each of the residues singly and in combination and assess them in uptake assays to verify data from the MD simulations. This is a whole new study and, given the uncertainties surrounding the observations of just superimposing TbAQP2 structure and the TbAQP3 model, we feel this is just too speculative to add to our manuscript.

(8) To validate the densities representing glycerol and the compounds, you should show halfmap densities for these.

A new figure, Fig S6 has been made to show the half-map densities for the glycerol and drugs.

(9) I would also like to see the density coverage of the individual helices/structural elements.

A new figure, Fig S5 has been made to show the densities for the structural elements.

(10) While the LigPlot figure is nice, I think showing the data (including the cryo-EM density) is necessary validation.

The LigPlot figure is a diagram (an interpretation of data) and does not need the densities as these have already been shown in Fig. 1c (the data).

(11) I would recommend including a figure that illustrates the points described in lines 123-134.

All of the points raised in this section are already shown in Fig. 2a, which was referred to twice in this section. We have added another reference to Fig.2a on lines 134-135 for completeness.

(12) Line 202: I would suggest using "membrane potential/voltage" to avoid confusion with mitochondrial membrane potential.

We have changed this to ‘plasma membrane potential’ to differentiate it from mitochondrial membrane potential.

| (13) *Figure 4: Label C.O.M. in the panels so that the figure corresponds to the legend.*

We have altered the figure and added an explanation in the figure legend (lines 716-717):

‘Cyan mesh shows the density of the molecule across the MD simulation, and the asterisk shows the position of the centre of mass (COM).’

| (14) *Figure S2: Panels d and e appear too similar, and it is difficult to see the stick representation of the compound. I would recommend either using different colours or showing a close-up of the site.*

We have clarified the figure by including two close-up views of the hot-spot region, one with melarsoprol overlaid and one with pentamidine overlaid

| (15) *Figure S2: Typo in legend: 8YJ7 should be 8JY7.*

Changed as suggested

| (16) *Figure S3 and Figure S4: Please clarify which parts of the process were performed in cryoSPARC and which in Relion.*

Figure S3 gives an overview of the processing and has been simplified to give the overall picture of the procedures. All of the details were included in the Methods section as other programmes are used, not just cryoSPARC and Relion. Given the complexities of the processing, we have referred the readers to the Methods section rather than giving confusing information in Fig. S3.

We have updated the figure legend to Fig. S4 as requested.

| (17) *Figure S9 and Figure S10: The legends are swapped in these two figures.*

The captions have been swapped to their proper positions.

| (18) *For ease of orientation and viewing, I would recommend showing a vertical HOLE plot aligned with an image of the AQP2 pore.*

The HOLE plot has been re-drawn as suggested (Fig. S2)

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